# Collective Purchasing under Private Information: Efficiency and Cost Recovery

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**Abstract**

We study collective purchasing in which a given group of users with privately known productivities bids through a mediator against outside bidders in a second-price auction for a fixed divisible resource sold as a single lot. Sharing can improve resource use and make additional purchases worthwhile, but private information may prevent efficient purchasing from recovering expected costs. Under a regularity condition, we characterise the surplus-maximising mechanism subject to truthful reporting, voluntary participation, and expected cost recovery. It uses adjusted productivity weights to choose the group's bid and internal allocation. When efficient purchasing cannot recover expected costs, the mechanism restores cost recovery with the smallest possible loss of expected surplus. Its bid is no higher than the efficient benchmark, so beneficial purchases may be forgone. We also identify conditions under which acquired capacity is weakly more concentrated among higher-productivity users, potentially reducing total use value. With independent productivity draws and fixed total capacity, Vickrey–Clarke–Groves payments implement the exact first best and recover expected costs for all sufficiently large finite groups. As group size tends to infinity, we also characterise the division of expected surplus between the mediator and users, relating their shares to the production function.

**Keywords:** Collective purchasing; mechanism design; cost recovery; second-price auction; shared capacity.

**JEL classification:** D82, D44, D71.

# 1 Introduction

In 2016, ten farms covering about 5,200 hectares in western Victoria were offered for sale as a single portfolio. Local farming families were interested in purchasing individual farms to expand their holdings, but the seller would not sell the farms separately. Chris Mulcahy, an accountant in Ballarat, saw an opportunity and approached farming clients he knew were looking for land. Seven families joined together and purchased the portfolio for approximately

A$30 million in a competitive sale. To facilitate the purchase, Mulcahy needed to determine which farms each family wanted, how much each was prepared to pay, and whether their combined contributions could support an offer for the entire portfolio (Rowley, 2016; Schlesinger, 2016).

To decide how much to offer for the portfolio, Mulcahy would need to know what the land would be worth in its members' hands. But farmers know more about these returns than he does. A farmer may have a reason to understate that value if doing so lowers her contribution, even though her report may also change the land she receives. Each farmer must also expect to benefit from joining. How can a buying group make efficient purchasing and allocation decisions while encouraging truthful reporting and voluntary participation, and still collect enough to pay for the entire portfolio?

Kamble and Walrand (2015) construct coalition-strategyproof mechanisms for a related collective purchasing problem that encourage truthful reporting using resource and payment shares specified in advance for each possible group subset. Their mechanisms recover the purchase price exactly whenever the group wins but may forgo purchases that another division of the resource and its cost would make acceptable to all members, leaving potential gains from trade unrealised.

We ask how a buying group should choose its bid, allocation, and payments to maximise expected surplus when acquisition costs need only be recovered on average.[1] To study it, we consider a fixed group seeking to acquire a resource sold as a single lot and divide it among its members. Unlike the ten distinct farms in Demeter, the resource in our model is homogeneous and divisible. Each user privately knows her productivity $\theta_i$ and derives value $\theta_i u(q_i)$ from receiving $q_i$ units, where $u$ is increasing and strictly concave. The Demeter syndicate faced competition from domestic and overseas investors. We represent this competition through a second-price auction in which outside buyers bid separately against the group. A mediator uses members' reports to determine the group bid and, if the group wins, their allocations and payments. We measure surplus as the group's total value from using the resource less its acquisition costs.

We first characterise when the group can finance the first-best purchasing and allocation decisions. If members' productivities were known, the group would allocate the resource to maximise total value and bid that amount. This is the first-best benchmark. With private information, however, members may need to keep some of the gains from purchasing together to make truthful reporting and participation worthwhile. These required gains, or *information rents*, can in expectation exceed the surplus created by collective purchasing, making the first best financially infeasible. Building on Krishna and Perry (1998), we show that, in such

[1]The optimisation is subject to Bayesian incentive compatibility and interim individual rationality.

cases, changing payments alone cannot solve this financing problem.

What should the group do instead? Dearden (1997) uses adjusted valuations to study efficient provision under financing constraints for an excludable public good. We apply this approach to a buying group that must divide a rival resource after acquiring it. Under regularity, the same adjusted productivity weights determine both the group's bid and its internal allocation. Choosing the smallest adjustment that recovers expected costs yields the mechanism with the least loss of expected surplus.

These second-best rules reduce information rents for the members but also reduce the value generated by the group. We identify two sources of such loss. First, it bids no higher than the first-best rule, so the group may lose a purchase even when it could generate higher production value than any other outside buyers. Second, even when both rules acquire the resource, the second-best allocation may differ from the first best. While Polishchuk and Tonis (2013) show that revenue maximisation shifts resources towards higher types, we show that such distortions may arise even when the objective is to maximise surplus subject to cost recovery. Under an additional distributional condition, we further establish that increasing the adjustment weakly raises the combined capacity share of the top $k$ productivity types, for every $k$ smaller than the total group size.

The Demeter syndicate brought together seven farming families. What if more farmers could share the same amount of acquired land? Birulin (2006) shows that efficiency and budget balance can be achieved for congestible public goods in large economy, but he allows capacity to grow with population. Since Demeter portfolio only involved a fix amount of land, we instead keep rival resource capacity fixed in our model. A larger group creates more opportunities to use the fixed resource productively, while making the group less dependent on any single farmer. As the group grows, the expected surplus can eventually be large enough to cover the amount of information rents. We show that, for all sufficiently large finite groups, the first-best purchasing and allocation decisions can be implemented while recovering costs on average. The group no longer needs to sacrifice efficiency to recover its costs.

Levy et al. (2015) also study large markets for a fixed divisible resource. They show that, with sufficiently many buyers, dividing the resource among multiple winners generates more seller revenue than selling it as a whole to a single winner. In Demeter, however, the seller required a whole-lot sale, leaving buyers to finance the purchase jointly. Our result addresses this financing problem for buyer side. Their focus on revenue also raises a related question: even when the group can finance the purchase, how much of the surplus do its members retain?

As the land is divided among more farmers, the value created by an additional farmer

depends on how productive small allocations of land remain. These additional gains also affect the benefits members must retain.[2] We show that the division of surplus in large groups depends on the production function's local elasticity $qu'(q)/u(q)$ at zero, which we denote the limit as $\rho$. Provided expected first-best surplus is positive for some finite group, the mediator's expected budget and members' aggregate benefits, as shares of expected first-best surplus, converge to $\rho$ and $1-\rho$, respectively. Thus, when $\rho < 1$, members may collectively retain a positive share of the gains.

The Demeter transaction also suggests a direction for future research. Mulcahy described the difficulty of getting everyone to agree to the purchase, and he was also a co-investor. Our model takes group membership as given and assumes that the mediator implements agreed rules without bearing search or coordination costs. Giving an organiser a financial stake may encourage the effort needed to assemble a buying group. But if the organiser can influence who joins or how the land and costs are divided, that stake may also affect these decisions.

A second question concerns whether the seller offers the properties together or separately. Unlike Demeter, the six properties in South Australia's Saltbush Ag aggregation were offered both as a single portfolio and as individual lots. They were eventually sold separately to five local family farming businesses (Ray White, 2023). Our model assumes that the group must purchase the entire resource and that its members cannot buy directly from the seller. When individual properties are available separately, however, farmers can pursue their own purchases without helping to finance the rest of the portfolio. This changes their incentives to join a syndicate and how much they are willing to contribute.

The remainder of the paper is set out as follows. Sections 2 and 3 present the model and design problem. Section 4-7 establishes the analysis of financing problem, optimal mechanisms, and group size. Section 8 concludes. The appendices contain the technical foundations and proofs.

# 2 Environment

**Resources and agents.** A seller auctions a concession to use a perfectly divisible resource with total capacity $\bar{Q} > 0$. There are two classes of potential buyers. One class consists of $n \geq 2$ users who participate jointly as a group through a mediator. The other consists of outside bidders who compete for the concession individually. The concession is awarded as a whole, but the capacity is divisible. Thus, if the group wins, the available capacity must be allocated within the group, and this allocation is coordinated through the mediator. Importantly, we take group membership as given and exclude trade between the group and

[2]These are normalised Vickrey–Clarke–Groves (VCG) payments.

outside bidders.

**External auction.** The seller uses a second-price auction. The mediator submits a single bid on behalf of the group, but individual users cannot bid separately. Each outside bidder submits an individual bid. The highest bidder wins the concession and pays the second-highest bid.

Outside bidders have private values and quasilinear preferences. The highest outside bid is denoted by $Y$. The distribution $F_Y$ of $Y$ has support $\mathcal{Y} \subseteq \mathbb{R}_+$, a continuous density $f_Y$, and finite mean. Thus, the group acquires the concession whenever its bid is weakly greater than $Y$, in which case it pays $Y$ to the seller.

**Users and shared capacity.** User $i$ privately observes a productivity type $\theta_i \in [0, \bar{\theta}]$, where $\bar{\theta} > 0$. Types are independently drawn from a common absolutely continuous distribution $H$, with continuous and strictly positive density $h$. They are independent of the outside bid $Y$. A type profile $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ records the productivities of all $n$ users.

If user $i$ receives capacity $q_i$, her use value is $\theta_i u(q_i)$. The common production function $u : [0, \bar{Q}] \to \mathbb{R}_+$ is continuous, twice continuously differentiable on $(0, \bar{Q}]$, strictly increasing, and strictly concave, with $u(0) = 0$. Strict concavity captures diminishing returns: the marginal benefit of capacity falls as a user receives more of it.

An allocation is feasible if each user receives a nonnegative amount of capacity and total allocated capacity does not exceed $\bar{Q}$:

$$\mathcal{Q} = \left\{ \boldsymbol{q} \in \mathbb{R}_+^n : \sum_{i=1}^{n} q_i \leq \bar{Q} \right\}.$$

The capacity may be divided among several users or assigned entirely to a single user.

**Internal mechanism and mediator.** An internal mechanism coordinates decisions within the group. The mediator collects a report $r_i \in [0, \bar{\theta}]$ from each user, which may differ from the user's true productivity. The mediator then implements three functions fixed in advance: a bid function $B$, an allocation function $\boldsymbol{q}$, and a payment function $\boldsymbol{p}$, collectively denoted by $\mathcal{M} = (B, \boldsymbol{q}, \boldsymbol{p})$. Given a report profile $\boldsymbol{r} = (r_1, \ldots, r_n)$, they determine the group bid $B(\boldsymbol{r}) \in \mathbb{R}_+$, the capacity allocation $\boldsymbol{q}(\boldsymbol{r}, Y) \in \mathcal{Q}$, and user payments $\boldsymbol{p}(\boldsymbol{r}, Y) \in \mathbb{R}^n$. The bid depends only on reports because it is submitted before $Y$ is observed. The allocations and payments may also depend on the realised outside bid.

If $B(\boldsymbol{r}) < Y$, the group loses and its users receive no capacity. We also require that no transfers take place between users and the mediator after a loss. Thus, $\boldsymbol{q}(\boldsymbol{r}, Y) = \mathbf{0}$ and

$\boldsymbol{p}(\boldsymbol{r}, Y) = \mathbf{0}$. A type-$\theta_i$ user's realised utility is her use value less her payment:

$$\theta_i u(q_i(\boldsymbol{r}, Y)) - p_i(\boldsymbol{r}, Y).$$

**Timing.** The mechanism $\mathcal{M}$ is fixed and announced before users observe their types. Users then observe their own types and report to the mediator without observing the other users' types or $Y$. The mediator submits $B(\boldsymbol{r})$ before $Y$ is observed. Once $Y$ is revealed, the auction outcome is determined. If the group wins, it pays $Y$ to the seller and the mediator implements the prescribed allocation $\boldsymbol{q}(\boldsymbol{r}, Y)$ and user payments $\boldsymbol{p}(\boldsymbol{r}, Y)$.

# 3 Design problem

We now ask how the coordination rules should be chosen. Taking the seller's second-price auction as given, a designer selects $\mathcal{M} = (B, \boldsymbol{q}, \boldsymbol{p})$ [3]. The mediator then implements the announced mechanism.

**Surplus objective.** The designer pursues the goal of maximising expected incremental surplus. This measures the additional value created by allocating the concession to the group rather than to an outside bidder. Truthful bidding is weakly dominant for outside bidders in the seller's second-price auction (Vickrey, 1961), so we take them to bid truthfully. The highest outside bid $Y$ is therefore also the value of the best outside use.

If the group wins, its users generate total value $\sum_{i=1}^{n} \theta_i u(q_i)$, while outside use worth $Y$ is forgone. Incremental surplus is the difference between these values. If the group loses, the concession remains with an outside bidder, so incremental surplus relative to this benchmark is zero. Evaluated at truthful reports $\boldsymbol{r} = \boldsymbol{\theta}$, expected incremental surplus is therefore

$$S(\mathcal{M}) = \mathbb{E}\left[\mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\}\left(\sum_{i=1}^{n} \theta_i u(q_i(\boldsymbol{\theta}, Y)) - Y\right)\right].$$ [4]

The expectation is over users' types and the outside bid.

**Incentive compatibility and participation.** A user's report affects the group's bid, her allocation, and her payment, so she may benefit from misreporting her productivity. The mechanism must therefore make truthful reporting optimal. Since she reports without

[3] We consider payment rules that are integrable under truthful reporting and have finite interim expected payments for every report.

[4] The indicator equals one when the group wins and zero otherwise. The expression inside the bracket is equivalent to $\sum_{i=1}^{n} \theta_i u(q_i(\boldsymbol{\theta}, Y)) - \mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\} Y$.

knowing the other users' types or $Y$, her incentives are evaluated in expectation over these unknown quantities. When the other users report truthfully, her interim expected utility from truthful reporting is

$$U_i(\theta_i) := \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[\theta_i u(q_i(\boldsymbol{\theta}, Y)) - p_i(\boldsymbol{\theta}, Y)]\,,$$

where $\boldsymbol{\theta}_{-i}$ denotes the other users' types. The term interim refers to evaluation after the user has learned her own type but before the remaining uncertainty is resolved.

**Definition 3.1.** A mechanism is *Bayesian incentive compatible* (BIC) if, for every $i$ and every $\theta_i, r_i \in [0, \bar{\theta}]$,

$$U_i(\theta_i) \geq \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[\theta_i u(q_i((r_i, \boldsymbol{\theta}_{-i}), Y)) - p_i((r_i, \boldsymbol{\theta}_{-i}), Y)]\,.$$

The right-hand side is the expected utility from reporting $r_i$ instead of the true type $\theta_i$, while the other users continue to report truthfully. Thus BIC requires truthful reporting to be optimal given the information available at the reporting stage.

Participation is evaluated at the same stage. A user who does not participate receives no capacity and obtains utility zero. The mechanism must therefore give every type at least this expected utility.

**Definition 3.2.** A mechanism is *interim individually rational* (interim IR) if $U_i(\theta_i) \geq 0$ for every user $i$ and every $\theta_i \in [0, \bar{\theta}]$.

A characterisation of BIC in terms of the allocation and payment rules is given in Proposition A.7 in Appendix A.3.

**Budget requirement.** The designer also requires expected user payments to cover the expected acquisition cost. We measure the mediator's budget as the sum of user payments less the group's payment to the seller. After a win, the group pays $Y$ to the seller. After a loss, no user payments are collected and no payment is made to the seller. Evaluated at truthful reports, the mediator's expected budget is therefore

$$\Pi(\mathcal{M}) = \mathbb{E}\left[\mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\}(\sum_{i=1}^{n} p_i(\boldsymbol{\theta}, Y) - Y)\right].^{5}$$

The expectation is over user types and the outside bid.

[5] The expression inside the bracket is equivalent to $\sum_{i=1}^{n} p_i(\boldsymbol{\theta}, Y) - \mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\}Y$.

**Definition 3.3.** The mechanism satisfies *expected cost recovery* if $\Pi(\mathcal{M}) \geq 0$. This condition is also called ex ante no deficit.

This condition allows the mediator's budget to be negative in some states, provided that these losses are offset in expectation by positive budgets in other states.

**Design problem.** Among the mechanisms described in Section 2, the designer therefore chooses the internal mechanism $\mathcal{M}$ to maximise expected incremental surplus subject to BIC, interim IR, and expected cost recovery:

$$\begin{aligned} \sup_{\mathcal{M}} \quad & S(\mathcal{M}) \\ \text{subject to} \quad & \mathcal{M} \text{ is BIC and interim IR, } \Pi(\mathcal{M}) \geq 0. \end{aligned} \tag{1}$$

# 4 First best and VCG deficits

Before solving the design problem, we first identify the group bid and internal allocation that maximise surplus when users' productivities are observed. We then introduce VCG payments to implement these first-best decisions under private information and examine whether they cover the acquisition cost.

## 4.1 First-best benchmark

First suppose that the designer observes all users' productivities and faces no budget constraint. Conditional on the group receiving the concession, the designer allocates capacity across users so as to maximise the group's total value. For any nonnegative productivity profile $\boldsymbol{\theta} \in \mathbb{R}^n_+$, define

$$W(\boldsymbol{\theta}) = \max_{\boldsymbol{q} \in \mathcal{Q}} \sum_{i=1}^{n} \theta_i u(q_i), \qquad \boldsymbol{q}^*(\boldsymbol{\theta}) \in \arg\max_{\boldsymbol{q} \in \mathcal{Q}} \sum_{i=1}^{n} \theta_i u(q_i). \tag{2}$$

Here $W(\boldsymbol{\theta})$ is the maximum total use value the group can generate, and $\boldsymbol{q}^*(\boldsymbol{\theta})$ is the allocation that achieves it. We set $\boldsymbol{q}^*(\mathbf{0}) = \mathbf{0}$.[6]

To see how the efficient allocation is determined, consider two users who both receive positive capacity. If one user has a higher marginal benefit, $\theta_i u'(q_i)$, than the other, shifting a small amount of capacity toward that user raises total value. Hence, at the efficient allocation, marginal benefits are equalised across users who receive positive capacity. Figure 1 illustrates this condition for a two-user group in which both users receive positive capacity.

[6] Lemmas A.1 and A.2 in Appendix A.1 establish the relevant properties of this allocation problem.

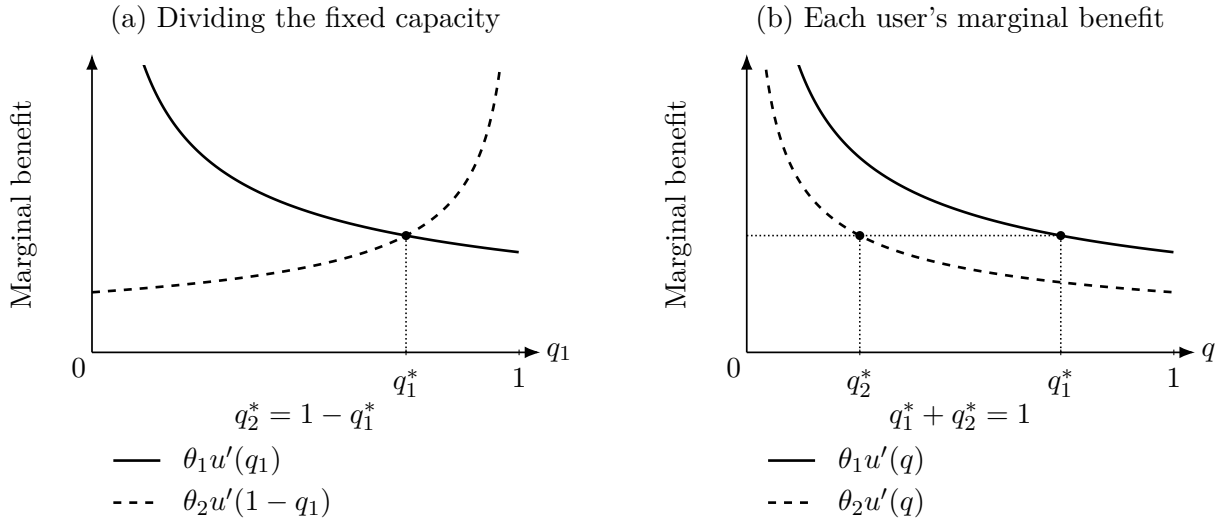



**Figure 1.** Two views of the same first-best allocation, conditional on acquisition. Capacity is normalised to $\bar{Q} = 1$. In the left panel, the horizontal axis is user 1's allocation, so user 2 receives $1 - q_1$. The right panel plots each user's marginal benefit against her own quantity. The common marginal-benefit level identifies two allocations whose sum is 1.

Having identified the group's best internal use, we next compare it with outside use, worth $Y$. Bidding $W(\boldsymbol{\theta})$ makes the group acquire whenever its maximum value is at least $Y$. If the group wins, capacity is allocated according to $\boldsymbol{q}^*(\boldsymbol{\theta})$. Otherwise, every user receives zero capacity. The first-best rules are therefore

$$B^\circ(\boldsymbol{\theta}) = W(\boldsymbol{\theta}), \qquad \boldsymbol{q}^\circ(\boldsymbol{\theta}, Y) = \mathbb{1}\{W(\boldsymbol{\theta}) \geq Y\}\boldsymbol{q}^*(\boldsymbol{\theta}). \tag{3}$$

A superscript $\circ$ denotes first-best objects.

A win creates incremental surplus $W(\boldsymbol{\theta}) - Y$ relative to outside use, while a loss creates zero incremental surplus. First-best expected incremental surplus is therefore

$$S^\circ = \mathbb{E}\Big[\mathbb{1}\{W(\boldsymbol{\theta}) \geq Y\}\Big(W(\boldsymbol{\theta}) - Y\Big)\Big].$$

At this stage, we have specified the group's bid and internal allocation, but not user payments.

## 4.2 VCG and its deficit

We introduce VCG payments to implement the first-best decisions under private information and illustrate why they may generate a realised deficit.

**VCG mechanism.** We first consider the Vickrey–Clarke–Groves (VCG) mechanism. It applies the first-best bid and allocation rules in (3) to reported productivities and adds a

VCG payment rule $\widetilde{p}_i(\boldsymbol{r}, Y)$ for each user. The payment compares the maximum use value available to the other users and outside bidders without user $i$ with the value they obtain when she is present.

If user $i$ were absent, the remaining users could still share the entire capacity. For reports $\boldsymbol{r}_{-i}$, define their maximum value by

$$W_{-i}(\boldsymbol{r}_{-i}) = \max_{\substack{\boldsymbol{q}_{-i} \in \mathbb{R}_+^{n-1} \\ \sum_{j \neq i} q_j \leq \bar{Q}}} \sum_{j \neq i} r_j u(q_j).$$

Without user $i$, the concession could go either to the remaining users or to an outside bidder. The highest reported use value available without her is therefore $\max\{Y, W_{-i}(\boldsymbol{r}_{-i})\}$.

If user $i$ is present and the group wins, outside bidders receive no concession, while the other users generate reported value $\sum_{j \neq i} r_j u(q_j^*(\boldsymbol{r}))$. The payment charges user $i$ the difference between these two values:

$$\widetilde{p}_i(\boldsymbol{r}, Y) = \mathbb{1}\{W(\boldsymbol{r}) \geq Y\} \left[ \max\{Y, W_{-i}(\boldsymbol{r}_{-i})\} - \sum_{j \neq i} r_j u(q_j^*(\boldsymbol{r})) \right]. \tag{4}$$

Rewriting the payment separates two effects:

$$\widetilde{p}_i(\boldsymbol{r}, Y) = \mathbb{1}\{W(\boldsymbol{r}) \geq Y\} \left[ \underbrace{W_{-i}(\boldsymbol{r}_{-i}) - \sum_{j \neq i} r_j u(q_j^*(\boldsymbol{r}))}_{\text{within-group externality}} + \underbrace{\max\{Y - W_{-i}(\boldsymbol{r}_{-i}), 0\}}_{\text{acquisition externality}} \right].$$

Conditional on acquisition, the payment has two components. The first term holds the group's access to the concession fixed. It compares the maximum value the other users could generate using all the capacity with the value they generate when sharing it with user $i$. The second term accounts for the alternative of outside use. It is zero if the remaining group's value is at least $Y$; otherwise, it equals the amount by which outside use exceeds the remaining group's best internal value.

With the payment rule in (4), truthful reporting remains optimal for every outside bid and every profile of the other users' reports, and truthful users receive nonnegative payoffs in every realised state. Thus, VCG is statewise dominant-strategy incentive compatible (statewise DSIC) and ex post individually rational (EPIR) (Clarke, 1971; Groves, 1973).[7] It therefore satisfies the incentive and participation requirements of the design problem.

[7] Statewise DSIC and EPIR are defined in Definitions A.3 and A.4 in Appendix A.2.

VCG also charges nothing at a zero report. We call this payment property normalisation.[8] Since a zero-productivity user has no use value and makes no payment under normalisation, her interim utility is $U_i(0) = 0$.[9]

**VCG deficit.** It remains to check the budget requirement. Under VCG, the mediator's realised budget is

$$\sum_{i=1}^{n} \widetilde{p}_i(\boldsymbol{\theta}, Y) - \mathbb{1}\{W(\boldsymbol{\theta}) \geq Y\}Y,$$

that is, total payments collected from users less the amount paid to the seller. The budget difficulty is particularly clear when the group can beat the outside bid only with all its members present. We call a user *pivotal* if removing her changes a group win into a loss under the first-best rule.

Under truthful reporting, user $i$'s realised utility is

$$\theta_i u(q_i^\circ(\boldsymbol{\theta}, Y)) - \widetilde{p}_i(\boldsymbol{\theta}, Y) = \max\{Y, W(\boldsymbol{\theta})\} - \max\{Y, W_{-i}(\boldsymbol{\theta}_{-i})\}. \tag{5}$$

Thus, each user retains her marginal contribution to first-best incremental surplus.

If $W(\boldsymbol{\theta}) > Y > \max_i W_{-i}(\boldsymbol{\theta}_{-i})$, every user is pivotal for acquisition. Removing any one of them destroys the same joint gain, $W(\boldsymbol{\theta}) - Y$. VCG evaluates these counterfactual removals separately and leaves each user that full gain as utility. Yet the group creates the surplus only once. Aggregate user utility is therefore $n[W(\boldsymbol{\theta}) - Y]$, leaving the mediator with a realised budget of $(n-1)(Y - W(\boldsymbol{\theta})) < 0$.

# 5 Optimal mechanism

A realised VCG deficit need not imply an expected deficit, since positive budgets in other states may offset it. We now ask when the first-best bid and allocation rules can recover expected costs.

## 5.1 Largest expected budget

To answer this question, we characterise the largest expected budget attainable under a given pair of bid and allocation rules. Fix a bid rule $B$ and an allocation rule $\boldsymbol{q}$ that can be implemented by some BIC payment rule. We call such a pair $(B, \boldsymbol{q})$ BIC-implementable.

[8] Formally, a payment rule $\boldsymbol{p}$ is *normalised* if $p_i((0, \boldsymbol{r}_{-i}), Y) = 0$ for every user $i$, every $\boldsymbol{r}_{-i} \in [0, \bar{\theta}]^{n-1}$, and every $Y \in \mathcal{Y}$. A mechanism is normalised if its payment rule is normalised.

[9] Normalisation is not imposed in (1). This property will be useful when we compare alternative payment rules in Section 5.

We hold these rules fixed for every report profile and outside bid, and compare the payment rules that also satisfy interim IR.

With the bid and allocation fixed, payments do not affect expected incremental surplus, which we write as $S(B, \boldsymbol{q})$. They determine how that surplus is divided between the users and the mediator. The following identity makes this relationship explicit. Recall that $U_i(\theta_i)$ is user $i$'s interim expected utility.

**Lemma 5.1.** *For every mechanism with integrable payments,*

$$\underbrace{\Pi(\mathcal{M})}_{\text{expected budget}} = \underbrace{S(B, \boldsymbol{q})}_{\text{expected surplus}} - \underbrace{\sum_{i=1}^{n} \mathbb{E}_{\theta_i}[U_i(\theta_i)]}_{\text{users' expected utilities}} .$$

The mediator can raise its expected budget only by reducing users' expected utilities. The question is therefore how far these utilities can be reduced while preserving truthful reporting and participation.

A high-productivity user can report a lower productivity and obtain the allocation and payment intended for that report. She values any positive allocation more highly than the lower type. This possibility limits how much the mechanism can charge her while preserving truthful reporting. The envelope formula (Myerson, 1981)[10] gives

$$U_i(\theta_i) - U_i(0) = \int_0^{\theta_i} \mathbb{E}_{\boldsymbol{\theta}_{-i}, Y} \left[ u(q_i((t, \boldsymbol{\theta}_{-i}), Y)) \right] dt$$

for every type. We call $U_i(\theta_i) - U_i(0)$ the user's *information rent*: the additional utility a type $\theta_i$ receives relative to a type-zero user under truthful reporting. For fixed bid and allocation rules, the envelope formula pins down these information rents, so they are determined by the acquisition and allocation decisions rather than by the choice of BIC payment rule. Different BIC payment rules can only change the baseline utility $U_i(0)$, which interim IR requires to be nonnegative.[11]

To calculate aggregate expected information rents, we first average each user's rent over her productivity:

$$\mathbb{E}_{\theta_i}[U_i(\theta_i) - U_i(0)] = \mathbb{E}_{\boldsymbol{\theta}, Y}\left[ \frac{1 - H(\theta_i)}{h(\theta_i)} u(q_i(\boldsymbol{\theta}, Y)) \right].$$

Each report level $t$ enters the rent calculation for all types above $t$, whose total probability is $1 - H(t)$. Rewriting this integral as an expectation over types divides that probability

[10] The proof is given in Appendix A.3.

[11] At type zero, $U_i(0) = -\mathbb{E}_{\boldsymbol{\theta}_{-i}, Y}[p_i((0, \boldsymbol{\theta}_{-i}), Y)]$, because the user has no use value.

by the density $h(t)$. The ratio $(1 - H(t))/h(t)$ therefore expresses the rent adjustment in productivity units.

Since interim IR requires $U_i(0) \geq 0$, subtracting these required rents from expected surplus gives an upper bound on the mediator's expected budget. Define

$$\bar{\Pi}(B, \boldsymbol{q}) := \underbrace{S(B, \boldsymbol{q})}_{\text{expected surplus}} - \underbrace{\sum_{i=1}^{n} \mathbb{E}_{\theta_i}[U_i(\theta_i) - U_i(0)]}_{\text{aggregate information rents}}. \tag{6}$$

Together with the equation in Lemma 5.1, this definition gives us an upper bound on the mediator's expected budget:

$$\Pi(\mathcal{M}) = \bar{\Pi}(B, \boldsymbol{q}) - \sum_i U_i(0) \leq \bar{\Pi}(B, \boldsymbol{q}).$$

The bound can be attained by normalised BIC payments, which give $U_i(0) = 0$ for every user.[12] We therefore call $\bar{\Pi}(B, \boldsymbol{q})$ the *largest expected budget*. For the first-best bid and allocation rules, write $\bar{\Pi}^\circ := \bar{\Pi}(B^\circ, \boldsymbol{q}^\circ)$. Normalised VCG gives every zero-productivity type zero utility and therefore attains $\bar{\Pi}^\circ$.

**Proposition 5.2.** *The first-best bid and allocation rules $(B^\circ, \boldsymbol{q}^\circ)$ admit a BIC implementation satisfying expected cost recovery and interim IR if and only if $\bar{\Pi}^\circ \geq 0$.*

Thus, when VCG runs an expected deficit, changing payments alone cannot eliminate it while keeping the first-best decisions fixed.

We next express the largest expected budget at the level of individual productivity types, which will help us compare alternative bid and allocation rules. Define the user's *virtual productivity* by

$$\phi(\theta) = \theta - \frac{1 - H(\theta)}{h(\theta)}.$$

The budget bound can then be written as

$$\bar{\Pi}(B, \boldsymbol{q}) = \mathbb{E}\left[\mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\} \left(\sum_i \phi(\theta_i) u(q_i(\boldsymbol{\theta}, Y)) - Y\right)\right]. \tag{7}$$

Virtual productivity is true productivity net of the information-rent adjustment. It is a weight used to calculate expected budget, not a change in actual productivity. In particular, $\phi(\theta_i)u(q_i)$ is not the user's payment in a given realisation.

[12] See Lemma B.1 in Appendix B.2.

## 5.2 Adjusted-productivity mechanism

Subsection 5.1 allows us to separate the choice of payments from the choice of the bid and allocation rules. For any BIC-implementable pair $(B, \boldsymbol{q})$, payments can attain $\bar{\Pi}(B, \boldsymbol{q})$ without changing expected surplus. The design problem (1) is therefore equivalent to

$$\begin{aligned} \sup_{(B,\boldsymbol{q})} \quad & S(B, \boldsymbol{q}) \\ \text{subject to} \quad & (B, \boldsymbol{q}) \text{ is BIC-implementable}, \quad \bar{\Pi}(B, \boldsymbol{q}) \geq 0. \end{aligned} \tag{8}$$

We now focus on the case $\bar{\Pi}^\circ < 0$, in which the first-best rules cannot recover expected costs because their required information rents exceed the surplus they create. Since payments alone cannot eliminate this deficit, we search over all alternative pairs $(B, \boldsymbol{q})$, balancing the surplus each pair creates against the information rents it requires.

**Weighted problem.** Following the weighted-surplus approach (Ledyard and Palfrey, 2007; Loertscher and Marx, 2017), we introduce a weight $\alpha \in [0, 1]$ on required information rents and define the objective

$$\begin{aligned} L(B, \boldsymbol{q}; \alpha) &:= S(B, \boldsymbol{q}) - \alpha \underbrace{\sum_{i=1}^{n} \mathbb{E}_{\theta_i}[U_i(\theta_i) - U_i(0)]}_{\text{information rents}} \\ &= S(B, \boldsymbol{q}) - \alpha \Big[ S(B, \boldsymbol{q}) - \bar{\Pi}(B, \boldsymbol{q}) \Big] = (1 - \alpha) S(B, \boldsymbol{q}) + \alpha \bar{\Pi}(B, \boldsymbol{q}). \end{aligned} \tag{9}$$

As the first line shows, the weight $\alpha$ controls the trade-off between the surplus created by the rules and the information rents they require. Using the largest-budget relation in (6), we can also express the same objective as a weighted average of expected surplus and the largest expected budget.

Reducing rent requirements may therefore improve the largest expected budget at the cost of lower expected surplus. We call a departure from the first-best bid rule, allocation rule, or both a *distortion*, and measure its cost by the associated surplus loss. The reduced design problem (8) seeks expected cost recovery with the smallest possible loss of expected surplus.

To study this trade-off, we fix a weight $\alpha$ and choose the bid and allocation rules that maximise $L(B, \boldsymbol{q}; \alpha)$ over BIC-implementable pairs, without imposing the budget constraint at this stage. For each $\alpha$, the resulting rules determine both expected surplus and the largest expected budget. We then vary $\alpha$ to find the smallest weight that achieves expected cost

recovery.

Using the virtual-productivity representation (7) in the weighted objective (9) gives

$$L(B, \boldsymbol{q}; \alpha) = \mathbb{E}\left[\mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\} \left(\sum_{i=1}^{n} \Big[(1-\alpha)\theta_i + \alpha\phi(\theta_i)\Big] u(q_i(\boldsymbol{\theta}, Y)) - Y\right)\right].$$

In this objective, each user's allocation is valued using $(1-\alpha)\theta_i + \alpha\phi(\theta_i)$ rather than her true productivity $\theta_i$. Define the user's *adjusted productivity* by

$$\psi(\theta; \alpha) = \theta - \alpha \frac{1 - H(\theta)}{h(\theta)} = (1-\alpha)\theta + \alpha\phi(\theta). \tag{10}$$

Adjusted productivity thus translates the trade-off between surplus and required information rents into the weights used to choose acquisition and internal allocation. We next use these adjusted productivities to derive the bid and allocation rules for each $\alpha$.

**Bid and allocation rules.** To do so, we temporarily set aside incentive compatibility and maximise the expression inside the expectation for each type profile and outside bid. Adjusted productivity can be negative. Since leaving capacity unused is feasible, assigning zero capacity avoids the loss that a negative weight would cause in this objective. We therefore truncate adjusted productivity at zero:

$$\widehat{\psi}(\theta; \alpha) = \max\{\psi(\theta; \alpha), 0\}, \qquad \widehat{\boldsymbol{\psi}}(\boldsymbol{r}; \alpha) = \Big(\widehat{\psi}(r_1; \alpha), \ldots, \widehat{\psi}(r_n; \alpha)\Big).$$

After truncation, the allocation problem has the same form as the first-best allocation problem (2), with truncated adjusted productivities replacing true productivities. We therefore use the same value function $W$ and allocation rule $\boldsymbol{q}^*$ with different inputs. For a fixed weight $\alpha$, the mediator submits

$$\widehat{B}(\boldsymbol{r}; \alpha) = W\Big(\widehat{\boldsymbol{\psi}}(\boldsymbol{r}; \alpha)\Big) = \max_{\boldsymbol{q} \in \mathcal{Q}} \sum_{i=1}^{n} \widehat{\psi}(r_i; \alpha) u(q_i) \tag{11}$$

and uses the allocation rule

$$\widehat{\boldsymbol{q}}(\boldsymbol{r}, Y; \alpha) = \mathbb{1}\{\widehat{B}(\boldsymbol{r}; \alpha) \geq Y\} \boldsymbol{q}^*\Big(\widehat{\boldsymbol{\psi}}(\boldsymbol{r}; \alpha)\Big). \tag{12}$$

The bid in (11) is the largest total adjusted value the group can generate, so the group acquires whenever this value is at least the outside bid $Y$. Conditional on acquisition, the mechanism equalises adjusted marginal benefits, $\widehat{\psi}(r_i; \alpha) u'(q_i)$, across users receiving positive

capacity.[13]

**Incentives and payments.** We now ask whether the rules in (11) and (12) can be supported by truthful reporting. For this, we impose the following regularity condition on the type distribution.

**Assumption 5.3.** The virtual productivity $\phi(\theta)$ is weakly increasing on $[0, \bar{\theta}]$.

Regularity ensures that adjusted productivity is weakly increasing in a user's report. It follows from Lemma A.2 that $u(\widehat{q}_i(\boldsymbol{r}, Y; \alpha))$ is weakly increasing in $r_i$. The statewise DSIC characterisation in Proposition A.6 then allows these rules to be implemented using the normalised payment rule

$$\widehat{p}_i(\boldsymbol{r}, Y; \alpha) = r_i u(\widehat{q}_i(\boldsymbol{r}, Y; \alpha)) - \int_0^{r_i} u(\widehat{q}_i((t, \boldsymbol{r}_{-i}), Y; \alpha))\, dt. \tag{13}$$

Under truthful reporting, the first term is the user's use value and the integral is the utility she retains. The payment collects the remainder while preserving truthful reporting. Normalisation gives $U_i(0) = 0$ for every user, so Lemma B.1 implies that these payments attain the largest expected budget compatible with $(\widehat{B}, \widehat{\boldsymbol{q}})$.

Let $\widehat{\mathcal{M}}(\alpha) = (\widehat{B}, \widehat{\boldsymbol{q}}, \widehat{\boldsymbol{p}})$ denote the resulting *adjusted-productivity mechanism* at weight $\alpha$. The next proposition establishes optimality of its bid and allocation rules and the properties of their payment implementation.

**Proposition 5.4.** *Under Assumption 5.3, for every $\alpha \in [0, 1]$, the rules $(\widehat{B}, \widehat{\boldsymbol{q}})$ maximise $L(B, \boldsymbol{q}; \alpha)$ over all BIC-implementable pairs. The payments $\widehat{\boldsymbol{p}}$ attain $\bar{\Pi}(\widehat{B}, \widehat{\boldsymbol{q}})$ and make $\widehat{\mathcal{M}}(\alpha)$ normalised, statewise DSIC, and EPIR. Moreover,*

$$L(\widehat{B}, \widehat{\boldsymbol{q}}; \alpha) = \mathbb{E}\Big[\max\Big\{W\Big(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha)\Big) - Y,\, 0\Big\}\Big]. \tag{14}$$

*Payments are nonnegative and integrable, and users who receive no capacity make no payment.*

The proposition therefore gives us a family of mechanisms, $\{\widehat{\mathcal{M}}(\alpha)\}_{\alpha \in [0,1]}$. Although the optimisation is over the broad class of BIC and interim-IR mechanisms, an optimum is attained by a mechanism satisfying the stronger statewise DSIC and EPIR requirements.

[13] This need not equalise their true marginal benefits, because the weights also account for required rents.

## 5.3 Optimal mechanism

It remains to choose the weight that eliminates the expected deficit with the smallest possible loss of expected surplus. For each $\alpha \in [0,1]$, the corresponding surplus and budget are

$$S(\alpha) = S\Big(\widehat{\mathcal{M}}(\alpha)\Big), \qquad \Pi(\alpha) = \Pi\Big(\widehat{\mathcal{M}}(\alpha)\Big).$$

The normalised payments attain the largest expected budget for each mechanism's bid and allocation rules. Thus $\Pi(\alpha) = \bar{\Pi}(\widehat{B}, \widehat{\boldsymbol{q}})$, with both rules evaluated at $\alpha$, and aggregate expected information rents equal $S(\alpha) - \Pi(\alpha)$. We compare all three quantities along the mechanism family.

We begin with the two endpoints. At $\alpha = 0$, adjusted productivity equals true productivity, so the mechanism reproduces the first-best rules with $S(0) = S^{\circ}$. Since we now focus on the case in which the first best is not feasible, $\Pi(0) = \bar{\Pi}^{\circ} < 0$.

At the other endpoint, $\alpha = 1$, adjusted productivity becomes virtual productivity $\psi(\theta; 1) = \phi(\theta)$. Write $\widehat{\phi}(\theta) = \max\{\phi(\theta), 0\}$ and $\widehat{\boldsymbol{\phi}}(\boldsymbol{\theta}) = \big(\widehat{\phi}(\theta_1), \ldots, \widehat{\phi}(\theta_n)\big)$. The weighted objective becomes $L(B, \boldsymbol{q}; 1) = \bar{\Pi}(B, \boldsymbol{q})$. Proposition 5.4 therefore shows that the rules at $\alpha = 1$ maximise the largest expected budget. Since the normalised payments attain this bound, $\widehat{\mathcal{M}}(1)$ maximises expected budget over all BIC and interim-IR mechanisms with integrable payments. Its budget is

$$\Pi(1) = \mathbb{E}\Big[\max\Big\{W\Big(\widehat{\boldsymbol{\phi}}(\boldsymbol{\theta})\Big) - Y, 0\Big\}\Big] \geq 0. \tag{15}$$

The inequality follows because the mechanism acquires only when the group's optimised virtual value is at least the outside bid $Y$. Thus the family always contains a mechanism with nonnegative expected budget.

We therefore have $\Pi(0) < 0 \leq \Pi(1)$. To select the smallest feasible adjustment, we next examine how $S(\alpha)$ and $\Pi(\alpha)$ vary with $\alpha$.

**Proposition 5.5.** *Under Assumption 5.3, $\Pi(\alpha)$ is weakly increasing and $S(\alpha)$ is weakly decreasing on $[0,1]$. Moreover, both functions are continuous on $[0,1]$.*

It follows that aggregate expected information rents, $S(\alpha) - \Pi(\alpha)$, are also continuous and weakly decreasing. A larger $\alpha$ therefore selects rules with weakly lower surplus and rents, but a weakly higher expected budget. Since the original problem requires only cost recovery, choose the smallest weight that meets this requirement:[14]

$$\alpha^* = \min\{\alpha \in [0,1] : \Pi(\alpha) \geq 0\}. \tag{16}$$

[14]The selection of the smallest budget-feasible adjustment follows the approach in Dearden (1997).

The set is nonempty because $\Pi(1) \geq 0$, and continuity ensures that the minimum exists. The following theorem shows that the corresponding mechanism $\widehat{\mathcal{M}}(\alpha^*)$ solves the original design problem, including the case in which the first best already recovers expected costs.

**Theorem 5.6.** *Under Assumption 5.3, $\widehat{\mathcal{M}}(\alpha^*)$, with $\alpha^*$ defined in (16), solves the design problem (1). More specifically:*

*(i) If $\bar{\Pi}^\circ \geq 0$, then $\alpha^* = 0$, and the first-best rules are optimal.*

*(ii) If $\bar{\Pi}^\circ < 0$, then $\alpha^* \in (0, 1)$ and $\Pi(\alpha^*) = 0$. The optimal mechanism therefore adjusts the first-best rules, and the budget constraint binds in expectation.*

This theorem brings the preceding analysis together. The largest-budget criterion determines whether the first-best decisions can satisfy expected cost recovery once the information rents required for truthful reporting are taken into account. If $\bar{\Pi}^\circ \geq 0$, no distortion is needed and the first-best rules solve the design problem. If $\bar{\Pi}^\circ < 0$, the first-best decisions are not feasible under the budget requirement. The adjusted-productivity mechanism selected by $\alpha^*$ then changes the acquisition and allocation rules so as to satisfy expected cost recovery while preserving as much expected surplus as possible.

Importantly, this optimality holds among all BIC and interim-IR mechanisms satisfying expected cost recovery, not merely within the adjusted-productivity family. In the deficit case, the largest expected budget for the selected rules is zero at $\alpha^*$. Any feasible alternative with higher expected surplus would have a nonnegative largest expected budget and, since $\alpha^* < 1$, a higher value of the same weighted objective. This contradicts Proposition 5.4, which establishes that the selected rules maximise that objective over all BIC-implementable bid and allocation rules.

The largest-budget relation explains how this adjustment restores cost recovery. As $\alpha$ increases, both expected surplus and aggregate information rents weakly decrease, but the budget rises whenever the reduction in rents exceeds the loss of surplus. Figure 2 illustrates this trade-off. Where information rents exceed surplus, expected budget is negative. When the first best runs a deficit, $\alpha^*$ is the first point at which surplus and rents coincide, so expected budget reaches zero. Further increases in $\alpha$ may raise the budget above what the design problem requires, but only at the cost of additional surplus.

However, information rents are utilities retained by users, not themselves an efficiency loss. We examine the sources of the surplus loss in Section 6.

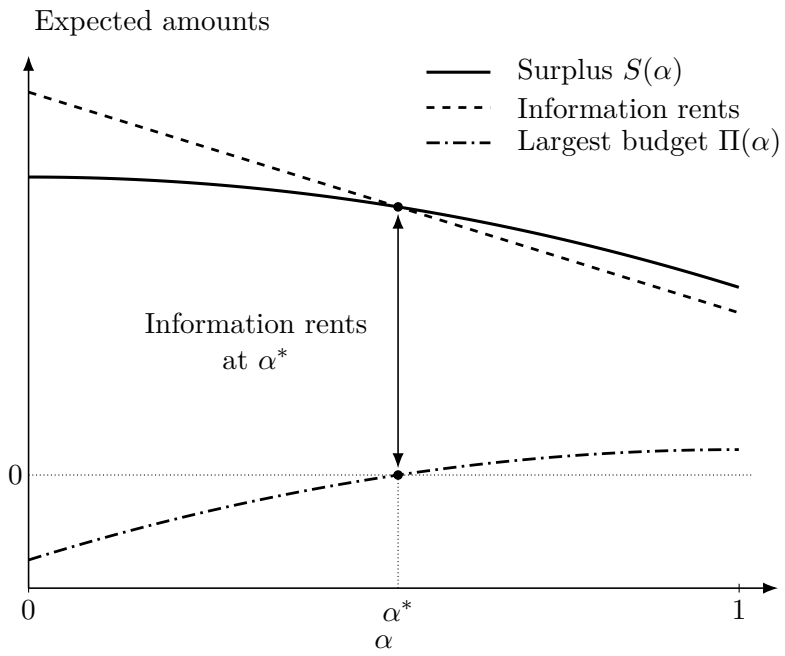



**Figure 2.** Surplus, information rents, and largest expected budget along an illustrative mechanism path with $\bar{\Pi}^\circ < 0$. All three curves use the same scale. The paths are schematic, not numerical model results.

# 6 Sources of efficiency loss

We now examine the decisions behind the loss of expected surplus: whether the group acquires the concession and how it uses the acquired capacity. Throughout, we maintain Assumption 5.3 and compare rules at the same true productivity profile and outside bid.

**Acquisition distortion.** Recall that the optimal mechanism replaces each true productivity by

$$\widehat{\psi}(\theta;\alpha^*) = \max\left\{\theta - \alpha^* \frac{1-H(\theta)}{h(\theta)}, 0\right\},$$

so the adjusted productivity is weakly lower than the true productivity. Since lowering any user's weight cannot raise the maximised value $W$, the group bid also weakly falls:

$$\widehat{B}(\boldsymbol{\theta};\alpha^*) = W(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha^*)) \leq W(\boldsymbol{\theta}) = B^\circ(\boldsymbol{\theta}).$$

Every state in which the cost-recovery mechanism acquires is therefore also a winning state under the first best. When $\widehat{B}(\boldsymbol{\theta};\alpha^*) < Y < W(\boldsymbol{\theta})$, the first best would acquire and create positive surplus $W(\boldsymbol{\theta}) - Y$, but the cost-recovery mechanism forgoes this purchase. The optimal mechanism therefore weakly shrinks the set of acquisitions and may prevent purchases that would generate positive surplus.

**Allocation distortion.** Even when both mechanisms acquire, they may divide the capacity differently. The first best allocates according to $\boldsymbol{q}^*(\boldsymbol{\theta})$, whereas the cost-recovery mechanism uses $\boldsymbol{q}^*(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha^*))$. In a state where both acquire, the surplus loss is

$$W(\boldsymbol{\theta}) - Y - [\sum_{i=1}^{n} \theta_i u\Big(q_i^*\Big(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha^*)\Big)\Big) - Y] = W(\boldsymbol{\theta}) - \sum_{i=1}^{n} \theta_i u\Big(q_i^*\Big(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha^*)\Big)\Big) \geq 0.$$

The inequality follows from first-best maximisation: efficiency depends on true productivities, not the adjusted weights used to choose the allocation. We next examine how this changes the distribution of capacity across users.

The bid depends on the level of optimised adjusted value $W(\widehat{\boldsymbol{\psi}})$, whereas the allocation conditional on acquisition depends on relative adjusted productivities. Multiplying all weights in a nonzero profile by the same positive factor scales the bid but leaves the allocation unchanged.[15] To explain changes in resource use, we must therefore compare proportional adjustments across users. For $\theta \in (0, \bar{\theta}]$, define the *relative rent wedge*

$$\chi(\theta) = \frac{\theta - \phi(\theta)}{\theta} = \frac{1 - H(\theta)}{\theta h(\theta)}. \tag{17}$$

Then adjusted productivity can be written as

$$\widehat{\psi}(\theta;\alpha) = \theta \max\{1 - \alpha\chi(\theta), 0\}.$$

For a positive adjusted weight, $\alpha\chi(\theta)$ is the proportional reduction from true productivity. Once this reduction reaches one, the weight is zero. To determine the direction of the allocation change, we impose an additional condition.

**Assumption 6.1.** The relative rent wedge $\chi$ is weakly decreasing on $(0, \bar{\theta}]$.[16]

Under this condition, lower-productivity types face weakly larger proportional reductions. The incentive link is the one established in Subsection 5.1: giving more capacity at low reports makes those reports more attractive to higher types, increasing the rents needed for truthful reporting. The relative rent wedge measures this rent adjustment relative to true productivity. A larger adjustment weight therefore makes lower-productivity uses relatively less attractive in the allocation problem. The first best already equalises true marginal benefits across users

[15] In particular, for any nonzero weight profile $\boldsymbol{w}$ and any $c > 0$, $W(c\boldsymbol{w}) = cW(\boldsymbol{w})$ and $\boldsymbol{q}^*(c\boldsymbol{w}) = \boldsymbol{q}^*(\boldsymbol{w})$.

[16] A weakly increasing hazard rate $h(\theta)/(1 - H(\theta))$ on $(0, \bar{\theta})$ is sufficient. Together with Assumption 5.3, this condition is the hyperregularity condition of Kleinberg and Yuan (2013, Definition 2.1). The condition is needed for the direction of the allocation comparison, not for the bid comparison or the nonnegativity of the surplus loss above.

receiving positive capacity. The adjusted rule instead equalises adjusted marginal benefits, so further concentration can reduce true use value.

The following theorem compares two members of the mechanism family from Subsection 5.2. Setting $\alpha_1 = 0$ and $\alpha_2 = \alpha^*$ gives the comparison between the first best and the optimal cost-recovery mechanism.

**Theorem 6.2.** *Suppose Assumptions 5.3 and 6.1 hold. Let $0 \leq \alpha_1 \leq \alpha_2 \leq 1$, fix a type profile, and relabel users so that $\theta_1 \geq \cdots \geq \theta_n$. Compare the allocations prescribed conditional on acquisition.*

*(i) The set of users receiving positive capacity weakly shrinks:*

$$\left\{i : q_i^*\left(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha_2)\right) > 0\right\} \subseteq \left\{i : q_i^*\left(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha_1)\right) > 0\right\}.$$

*(ii) If $\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha_2) \neq \mathbf{0}$, the total capacity assigned to the top $k$ productivity types weakly increases:*

$$\sum_{i=1}^{k} q_i^*\left(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha_2)\right) \geq \sum_{i=1}^{k} q_i^*\left(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha_1)\right), \quad k = 1, \ldots, n-1. \tag{18}$$

Thus, when the optimal cost-recovery allocation is nonzero, it assigns weakly more capacity to the top $k$ users jointly than the first best does, for every $k$. This is a cumulative comparison: not every member of that group must receive more. Concentration can increase without excluding anyone; in other cases, some previously served users receive no capacity. These are conditional allocations. If the group does not buy, every user's actual allocation is zero. That is the acquisition effect described above.

Taken together, the acquisition and allocation distortions account for the loss of expected surplus when averaged across states. As Section 5 showed, the optimal adjustment reduces aggregate expected information rents enough to offset this loss and eliminate the expected deficit.

# 7 Large groups

We now ask whether increasing group size allows the first-best bid and allocation rules to be implemented with a nonnegative expected budget, and how normalised VCG divides expected first-best surplus.

We increase the number of users to $m \geq 1$, while holding the production function $u$, total capacity $\bar{Q}$, type distribution $H$, and outside-bid distribution $F_Y$ fixed. Users'

types remain independent draws from $H$, independent of $Y$.[17] For the $m$-user economy, $W_m(\boldsymbol{\theta}) = \max \sum_{i=1}^{m} \theta_i u(q_i)$ is the maximum use value of the fixed capacity, and $S_m^\circ = \mathbb{E}[\max\{W_m(\boldsymbol{\theta}) - Y, 0\}]$ is expected incremental surplus under efficient acquisition, with $S_0^\circ = 0$. $\bar{\Pi}_m^\circ$ is the largest expected budget compatible with the first-best rules under BIC and interim IR. Normalised VCG attains this budget by Proposition 5.2.

## 7.1 The first best in large groups

Adding users can increase group use value through sharing. It also changes the aggregate expected information rents required to implement the first-best rules. To assess expected cost recovery, we compare these rents with expected first-best surplus.

To calculate expected information rents, recall that under normalised VCG, user $i$'s realised utility equals her marginal contribution to first-best incremental surplus (see (5)):

$$\max\{W_m(\boldsymbol{\theta}) - Y, 0\} - \max\{W_{m,-i}(\boldsymbol{\theta}_{-i}) - Y, 0\}.$$

That is, her utility is the first-best surplus created with her present minus the surplus that the remaining users could create without her, after reallocating the entire capacity and facing the same outside bid.

Taking expectations gives a particularly simple expression. Because users' types are i.i.d., removing any fixed user leaves $m-1$ independent draws from $H$. The expected first-best surplus of the remaining group is therefore $S_{m-1}^\circ$, while the expected surplus with all $m$ users is $S_m^\circ$. Hence each user's ex ante expected VCG utility is $S_m^\circ - S_{m-1}^\circ$. Normalisation gives $U_i(0) = 0$, so this is also each user's expected information rent. Aggregate expected information rents are therefore $m\big(S_m^\circ - S_{m-1}^\circ\big)$. Substituting this into the largest-budget relation gives the following financing identity.

**Proposition 7.1.** *For every $m \geq 1$,*

$$\bar{\Pi}_m^\circ = \underbrace{S_m^\circ}_{\text{expected surplus}} - \underbrace{m\big(S_m^\circ - S_{m-1}^\circ\big)}_{\text{aggregate expected information rents}}. \tag{19}$$

*In particular, $\bar{\Pi}_1^\circ = 0$. For $m \geq 2$,*

$$\bar{\Pi}_m^\circ \geq 0 \iff \underbrace{S_m^\circ - S_{m-1}^\circ}_{\text{expected rent per user}} \leq \underbrace{\frac{S_m^\circ}{m}}_{\text{average surplus per user}}. \tag{20}$$

[17] The analysis only uses the baseline assumptions. The additional distributional conditions in Assumptions 5.3 and 6.1 are not needed.

Each user retains her expected marginal contribution under normalised VCG. Cost recovery is possible exactly when this contribution is no greater than average first-best surplus per user.[18] Total surplus can continue to rise even when this condition holds.

This characterisation does not imply that the first-best expected budget improves with every additional user. As the group grows, diminishing returns reduce the expected marginal gain in group use value. But acquisition introduces a threshold effect. If $W_{m,-i} < Y < W_m$, user $i$ is pivotal: the group acquires with her but not without her, so removing her eliminates the entire realised surplus $W_m - Y$. Several users can be pivotal for the same surplus. Aggregate information rents can therefore remain high relative to surplus, even when the marginal gain in group use value is falling. The first-best budget may consequently worsen at some group sizes. Nevertheless, the next theorem shows that cost recovery is eventually achieved.

**Theorem 7.2.** *For all sufficiently large group sizes $m$, normalised VCG has nonnegative expected budget $\bar{\Pi}^{\circ}_m$ and therefore solves the $m$-user design problem.*

No distortion of acquisition or internal allocation is needed once the group is sufficiently large. This is exact attainment of the first best for all sufficiently large finite groups, not merely convergence to it. The economic explanation depends on the behaviour of $u$ near zero. With total capacity fixed, small allocations become important as the group grows.

**Finite marginal benefit at zero.** If $u'(0+)$ is finite, even a highest-productivity user has a bounded marginal benefit from the first units of capacity. Bounded productivities and fixed total capacity therefore place a finite upper bound on group value $W_m(\boldsymbol{\theta})$.[19]

As the group grows, each user becomes easier to replace by the remaining members, so the expected marginal gain in group use value $\mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]$ diminishes. A member's contribution to first-best surplus cannot exceed her contribution to group use value.[20] These expected marginal gains decrease and have a bounded cumulative total. They therefore become small enough for aggregate information rents to vanish, despite the growing number of users. If some finite group generates positive first-best surplus, larger groups continue to generate at least that surplus, while aggregate rents converge to zero. The first-best budget is therefore positive for all sufficiently large groups. If no group generates positive surplus,

[18] For $m \geq 2$, the condition is equivalently $S^{\circ}_m/m \leq S^{\circ}_{m-1}/(m-1)$: adding the $m$-th user does not increase surplus per user.

[19] The finite upper bound is $W_m(\boldsymbol{\theta}) \leq \bar{\theta} u'(0+) \bar{Q}$ for every group size $m$.

[20] In fact, these marginal contributions become small sufficiently quickly that

$$0 \leq m\big(S^{\circ}_m - S^{\circ}_{m-1}\big) \leq m\big(\mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]\big) \to 0.$$

both surplus and rents are zero, and the budget is zero as well. Figure 3 illustrates the positive-surplus case.

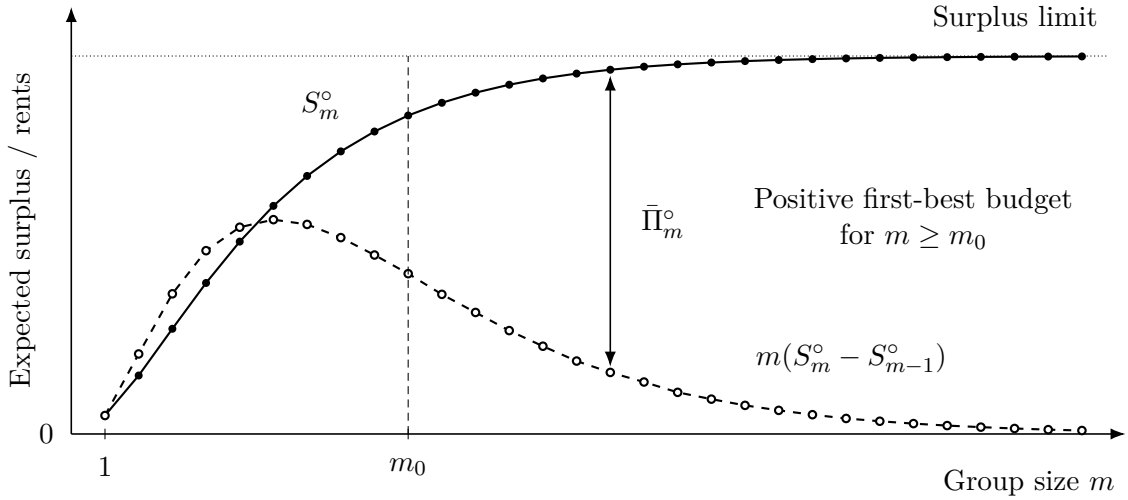



**Figure 3.** Schematic illustration of the finite-slope case $u'(0+) < \infty$ with positive limiting surplus. First-best surplus $S^\circ_m$ (solid) approaches a positive finite limit, while aggregate expected information rents $m(S^\circ_m - S^\circ_{m-1})$ (dashed) vanish as the group grows. Their vertical difference is the largest expected budget compatible with the first-best rules, $\bar{\Pi}^\circ_m$. The threshold $m_0$ shown in the figure is illustrative only. Points represent integer group sizes, and the paths are schematic.

**Unbounded marginal benefit near zero.** If $u'(0+) = \infty$, the marginal benefit of capacity becomes arbitrarily large as a user's allocation approaches zero. Adding users therefore creates new opportunities to exploit high marginal benefits at small allocations: the same fixed capacity can be shared among more users. As the group grows, this sharing effect makes expected group use value grow without bound, $\mathbb{E}[W_m] \to \infty$, even though total capacity remains fixed.

However, growing use value alone does not guarantee cost recovery, because truthful reporting requires information rents. Under normalised VCG, each user retains her marginal contribution to first-best incremental surplus. Since the remaining users can reallocate the capacity without her, this contribution cannot exceed her contribution to group use value. Aggregate expected information rents are therefore bounded by the group size multiplied by the expected gain in group use value from adding one user.

As the group grows, these expected marginal gains diminish, while total group use value continues to increase. Our analysis shows that expected group use value exceeds this upper bound on aggregate rents by an amount that grows without bound. Since expected first-best surplus is at least $\mathbb{E}[W_m] - \mathbb{E}[Y]$, where $\mathbb{E}[Y]$ is fixed and finite, the mediator's expected budget under normalised VCG also tends to infinity. The first-best rules therefore satisfy

expected cost recovery for all sufficiently large finite groups.

## 7.2 Surplus division in large groups

With finite marginal benefit at zero, aggregate expected information rents vanish as the group grows. However, this need not hold when marginal benefit near zero is unbounded. Eventual cost recovery only tells us when the first best can be sustained. Even if each user's expected rent becomes small as the group grows, aggregate rents need not be small because the number of users is also increasing. We therefore ask how normalised VCG divides first-best surplus between users' information rents and the mediator's expected budget.

We focus on the case $u'(0+) = \infty$. The key is the use value that the remaining users can recover by reallocating a user's capacity. As the group grows, the probability that it would still acquire after the removal of any single user tends to one. In these states, no member is pivotal for acquisition. A user's VCG payment then equals her within-group externality: the additional use value that the remaining users could generate by reallocating her capacity. Her realised utility is her use value less this payment, and hence her marginal contribution to first-best incremental surplus.

As the group grows, even the largest individual capacity share tends to zero (Lemma D.3(i)). A departing user's share can therefore be absorbed through small increases in the remaining allocations. Figure 4 illustrates this adjustment.

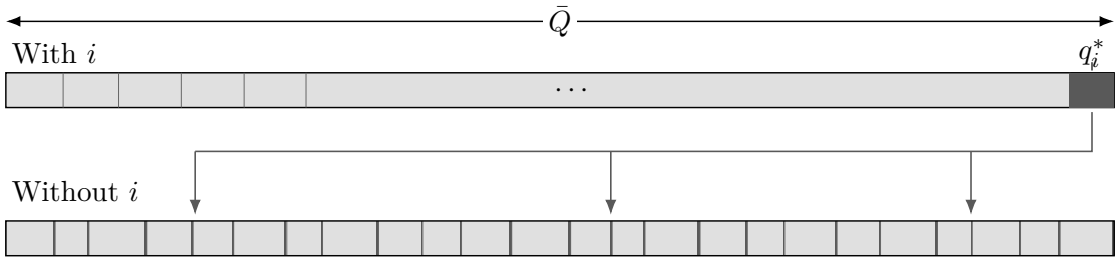



**Figure 4.** Reallocating a small share in a large group. Both bars represent the same total capacity $\bar{Q}$. The dark share $q_i^*$ in the upper bar is absorbed through many small increases in the remaining allocations below. The lower bar illustrates a feasible proportional redistribution.

The first-best allocation equalises marginal benefits across users receiving positive capacity. When a user's small allocation is reallocated, these marginal benefits help explain the additional use value that the remaining users can generate. Valuing the released capacity at the common marginal benefit gives

$$\underbrace{q_i^* \theta_i u'(q_i^*)}_{\text{capacity} \times \text{marginal benefit}} = \underbrace{\theta_i u(q_i^*)}_{\text{use value}} \times \underbrace{\frac{q_i^* u'(q_i^*)}{u(q_i^*)}}_{\text{local elasticity}} .$$

The local elasticity measures the proportional response of use value to a small proportional change in capacity. It relates a user's use value to the marginal benefit of her allocation, and thus helps explain the use value that the remaining users can recover by reallocation. Suppose this elasticity converges to $\rho$ as the allocation approaches zero. The reallocation argument then explains why aggregate VCG payments account for a limiting share $\rho$ of group use value, while users retain the complementary share $1-\rho$ as information rents. The next theorem further gives the limiting shares of expected first-best surplus after accounting for acquisition expenditure.

**Theorem 7.3.** *Suppose $qu'(q)/u(q) \to \rho \in [0,1]$ as $q \to 0$, and $S^\circ_m > 0$ for at least one group size. Under normalised VCG, as $m \to \infty$,*

$$\frac{\bar{\Pi}^\circ_m}{S^\circ_m} \longrightarrow \rho, \qquad \frac{m(S^\circ_m - S^\circ_{m-1})}{S^\circ_m} \longrightarrow 1-\rho.$$

*In words, the mediator's expected budget accounts for $\rho$ of expected first-best surplus. Users' expected information rents account for the complementary share $1-\rho$.*

A higher limiting elasticity gives the mediator a larger surplus share and leaves users a smaller aggregate rent share. If $u'(0+)$ is finite, the elasticity limit is one, and aggregate rents vanish as described in Subsection 7.1. With unbounded initial marginal benefit, acquisition expenditure becomes small relative to the group's growing use value, so it does not affect the limiting shares.

The shares compare expected total amounts under normalised VCG. A small share need not mean a small amount: when $\rho = 0$, the mediator's expected budget grows without bound even though its share tends to zero.[21] For $0 < \rho < 1$, users and the mediator both retain positive limiting shares.

# 8 Concluding remarks

This paper shows that gains from sharing do not guarantee that efficient collective purchasing can recover expected acquisition costs. Acquisition and internal allocation determine both expected surplus and the information rents required for truthful reporting. Cost recovery may therefore require changes in these decisions, rather than payments alone, while preserving truthful reporting and voluntary participation.

For a fixed group, we show how to recover expected costs with the smallest possible loss of expected surplus. Under regularity, the optimal mechanism uses adjusted productivity

[21] The case $\rho = 0$ requires $u'(0+) = \infty$, so the budget conclusion follows from the unbounded-marginal-benefit case above.

weights. Relative to the first best, its weakly lower bid can prevent beneficial purchases. Conditional on acquisition under both mechanisms, and under an additional distributional condition, internal allocation is weakly more concentrated among higher-productivity users at the same realised productivities.

Increasing group size can remove the need to sacrifice efficiency for cost recovery. We add users with independent productivity draws while holding total capacity, the production function, and the productivity and outside-bid distributions fixed. Normalised VCG then implements the exact first best with a nonnegative expected budget for all sufficiently large finite groups.

Restoring efficiency need not make aggregate information rents negligible. Suppose local elasticity converges as allocation approaches zero and expected first-best surplus is positive for some finite group. Under normalised VCG, the ratio of the mediator's expected budget to expected first-best surplus converges to this elasticity limit as the group grows. Users' aggregate expected information rents take the complementary limiting share.

# A Technical results

## A.1 The allocation problem

**Lemma A.1.** *For every $\boldsymbol{\theta} \neq \mathbf{0}$, the solution $\boldsymbol{q}^*(\boldsymbol{\theta})$ is unique and exhausts the available capacity,*

$$\sum_{i=1}^{n} q_i^*(\boldsymbol{\theta}) = \bar{Q}.$$

*Moreover, there exists a unique shadow value $\lambda(\boldsymbol{\theta}) \geq 0$ such that*

$$\begin{aligned} \theta_i u'(q_i^*(\boldsymbol{\theta})) &= \lambda(\boldsymbol{\theta}) && \text{if } q_i^*(\boldsymbol{\theta}) > 0, \\ \theta_i u'(0+) &\leq \lambda(\boldsymbol{\theta}) && \text{if } \theta_i > 0 \text{ and } q_i^*(\boldsymbol{\theta}) = 0, \\ q_i^*(\boldsymbol{\theta}) &= 0 && \text{if } \theta_i = 0. \end{aligned}$$

*The multiplier can equal zero only if $u'(\bar{Q}) = 0$ and exactly one component of $\boldsymbol{\theta}$ is positive. If $u'(0+) = +\infty$, every user with $\theta_i > 0$ receives positive capacity. If $u'(0+) < +\infty$, exclusion may occur.*

*Proof. Existence and binding.* The feasible set $\mathcal{Q}$ is nonempty and compact, and $\sum_i \theta_i u(q_i)$ is continuous, so an optimiser exists. Since some $\theta_i > 0$ and $u$ is strictly increasing, every optimiser exhausts capacity. A zero-type user cannot receive positive capacity, since transferring it to a positive-type user would strictly raise value. Thus $q_i^*(\boldsymbol{\theta}) = 0$ if $\theta_i = 0$.

*KKT characterisation.* Write $q_i^* = q_i^*(\boldsymbol{\theta})$. For $q_i^*, q_j^* > 0$, sufficiently small transfers in both directions are feasible. Optimality gives

$$\frac{d}{d\varepsilon}\Big[\theta_i u(q_i^* + \varepsilon) + \theta_j u(q_j^* - \varepsilon)\Big]\Big|_{\varepsilon=0} = \theta_i u'(q_i^*) - \theta_j u'(q_j^*) = 0.$$

Let $\lambda$ be this common marginal benefit, or the sole recipient's marginal benefit if only one user receives capacity. Then $\theta_i u'(q_i^*) = \lambda$ whenever $q_i^* > 0$, and $0 \le \lambda < \infty$.

If $\theta_i > 0$ but $q_i^* = 0$, choose $j$ with $q_j^* > 0$. Transferring $0 < \varepsilon < q_j^*$ from $j$ to $i$ is feasible. Optimality, $u(0) = 0$, and division by $\varepsilon$ give

$$\theta_i u(\varepsilon) + \theta_j[u(q_j^* - \varepsilon) - u(q_j^*)] \le 0,$$
$$\theta_i \frac{u(\varepsilon)}{\varepsilon} \le \theta_j \frac{u(q_j^*) - u(q_j^* - \varepsilon)}{\varepsilon}.$$

As $\varepsilon \to 0$, the right-hand side of the second inequality tends to $\theta_j u'(q_j^*) = \lambda$. Concavity and $u(0) = 0$ give $u(\varepsilon)/\varepsilon \to u'(0+)$, possibly infinite, so $\theta_i u'(0+) \le \lambda$.

Since $u'$ is nonnegative and strictly decreasing, it is positive on $(0, \bar{Q})$. Thus $\lambda = 0$ requires every user receiving capacity to receive $\bar{Q}$, with $u'(\bar{Q}) = 0$. Only one user can do so, and the boundary inequality rules out any other positive type.

*Uniqueness.* The objective is strictly concave in the coordinates with $\theta_i > 0$, and the feasible set is convex. All other coordinates are zero at an optimum, so the allocation is unique. Its common marginal benefit fixes $\lambda$ uniquely.

*Active users.* If $u'(0+) = +\infty$, the boundary inequality cannot hold for $\theta_i > 0$, because $\lambda < \infty$. Hence every positive type receives capacity. With finite $u'(0+)$, the inequality permits exclusion. □

**Lemma A.2.** *The optimised value $W$ is continuous, componentwise nondecreasing, convex, and positively homogeneous of degree one. The selected optimiser $\boldsymbol{q}^*(\boldsymbol{\theta})$ is continuous on $\mathbb{R}^n_+ \setminus \{\mathbf{0}\}$. Moreover, holding $\boldsymbol{\theta}_{-i}$ fixed,*

$$\theta_i' \ge \theta_i \quad \to \quad q_i^*(\theta_i', \boldsymbol{\theta}_{-i}) \ge q_i^*(\theta_i, \boldsymbol{\theta}_{-i}).$$

*Proof. Properties of the optimised value.* Because $\boldsymbol{q}^*(\boldsymbol{\theta}')$ is feasible at $\boldsymbol{\theta}$,

$$W(\boldsymbol{\theta}') - W(\boldsymbol{\theta}) \le \sum_{j=1}^n (\theta_j' - \theta_j) u\Big(q_j^*(\boldsymbol{\theta}')\Big) \le u(\bar{Q}) \sum_{j=1}^n |\theta_j' - \theta_j|.$$

The last step uses $0 \le u(q_j^*) \le u(\bar{Q})$. Interchanging the profiles gives

$$|W(\boldsymbol{\theta}') - W(\boldsymbol{\theta})| \le u(\bar{Q}) \sum_{j=1}^{n} |\theta_j' - \theta_j|,$$

so $W$ is Lipschitz and continuous. Increasing any weight weakly raises the objective at every feasible allocation, so $W$ is componentwise nondecreasing.

For $\tau \in [0,1]$, linearity in the weights gives

$$\begin{aligned} W(\tau\boldsymbol{\theta} + (1-\tau)\boldsymbol{\theta}') &= \max_{\boldsymbol{q}\in\mathcal{Q}} \left[\tau \sum_j \theta_j u(q_j) + (1-\tau) \sum_j \theta_j' u(q_j)\right] \\ &\le \tau W(\boldsymbol{\theta}) + (1-\tau) W(\boldsymbol{\theta}'). \end{aligned}$$

Thus $W$ is convex. For $c \ge 0$,

$$W(c\boldsymbol{\theta}) = \max_{\boldsymbol{q}\in\mathcal{Q}} c \sum_j \theta_j u(q_j) = cW(\boldsymbol{\theta}),$$

which proves positive homogeneity.

*Continuity of the optimiser.* The feasible set is fixed, nonempty and compact, and the objective is jointly continuous in weights and allocations. The optimiser is unique away from $\mathbf{0}$ by Lemma A.1. Berge's maximum theorem therefore gives continuity of $\boldsymbol{q}^*$ on $\mathbb{R}_+^n \setminus \{\mathbf{0}\}$.

*Monotonicity in weight.* Hold $\boldsymbol{\theta}_{-i}$ fixed. Equal own weights give the same selected allocation, including at $\mathbf{0}$, where $\boldsymbol{q}^*(\mathbf{0}) = \mathbf{0}$. For $\theta_i' > \theta_i$, write $\boldsymbol{q} = \boldsymbol{q}^*(\theta_i, \boldsymbol{\theta}_{-i})$ and $\boldsymbol{q}' = \boldsymbol{q}^*(\theta_i', \boldsymbol{\theta}_{-i})$. Each allocation is feasible at either weight, so the two optimality comparisons give

$$\begin{aligned} \theta_i u(q_i) + \sum_{j\ne i} \theta_j u(q_j) &\ge \theta_i u(q_i') + \sum_{j\ne i} \theta_j u(q_j'), \\ \theta_i' u(q_i') + \sum_{j\ne i} \theta_j u(q_j') &\ge \theta_i' u(q_i) + \sum_{j\ne i} \theta_j u(q_j). \end{aligned}$$

Adding the two inequalities cancels the other users' terms and gives $(\theta_i' - \theta_i)[u(q_i') - u(q_i)] \ge 0$. Since $\theta_i' - \theta_i > 0$ and $u$ is strictly increasing, $q_i' \ge q_i$. □

## A.2 Statewise properties and DSIC characterisation

We first define the stronger incentive, participation, and budget properties used to describe particular mechanisms in the main text. These properties are not imposed in (1).

**Definition A.3.** A mechanism is *statewise dominant-strategy incentive compatible* (statewise DSIC) if, for every user $i$, every $\theta_i, r_i \in [0, \bar{\theta}]$, every $\boldsymbol{r}_{-i} \in [0, \bar{\theta}]^{n-1}$, and every $Y \in \mathcal{Y}$,

$$\theta_i u(q_i((\theta_i, \boldsymbol{r}_{-i}), Y)) - p_i((\theta_i, \boldsymbol{r}_{-i}), Y) \geq \theta_i u(q_i((r_i, \boldsymbol{r}_{-i}), Y)) - p_i((r_i, \boldsymbol{r}_{-i}), Y).$$

Statewise DSIC requires truthful reporting to remain optimal for every outside bid and every profile of the other users' reports, whether or not those reports are truthful.

**Definition A.4.** A mechanism is *ex post individually rational* (EPIR) if, under truthful reporting,

$$\theta_i u(q_i(\boldsymbol{\theta}, Y)) - p_i(\boldsymbol{\theta}, Y) \geq 0$$

for every user $i$, type profile $\boldsymbol{\theta}$, and outside bid $Y \in \mathcal{Y}$.

EPIR requires participation to give nonnegative utility in every realised state, not merely in expectation after learning one's type.

**Definition A.5.** A mechanism satisfies *ex post cost recovery* if

$$\sum_{i=1}^{n} p_i(\boldsymbol{r}, Y) - \mathbb{1}\{B(\boldsymbol{r}) \geq Y\} Y \geq 0$$

for every report profile $\boldsymbol{r}$ and outside bid $Y \in \mathcal{Y}$.

Ex post no deficit requires user payments to cover the seller's price in every winning outcome.

Statewise DSIC implies BIC, EPIR implies interim IR, and ex post cost recovery implies ex ante cost recovery. We now characterise statewise DSIC in terms of allocation and payment rules.

**Proposition A.6.** *A mechanism is statewise DSIC if and only if, for every $i$, $\boldsymbol{r}_{-i}$, and $Y$, $u(q_i((r_i, \boldsymbol{r}_{-i}), Y))$ is nondecreasing in $r_i$ and*

$$p_i((r_i, \boldsymbol{r}_{-i}), Y) = r_i u(q_i((r_i, \boldsymbol{r}_{-i}), Y)) - \int_0^{r_i} u(q_i((t, \boldsymbol{r}_{-i}), Y))\, dt + p_i((0, \boldsymbol{r}_{-i}), Y).$$

*Under truthful reporting, user $i$'s realised utility is*

$$\theta_i u(q_i((\theta_i, \boldsymbol{r}_{-i}), Y)) - p_i((\theta_i, \boldsymbol{r}_{-i}), Y) = -p_i((0, \boldsymbol{r}_{-i}), Y) + \int_0^{\theta_i} u(q_i((t, \boldsymbol{r}_{-i}), Y))\, dt.$$

*Hence, under statewise DSIC, EPIR is equivalent to $p_i((0, \boldsymbol{r}_{-i}), Y) \leq 0$ for every $i$, $\boldsymbol{r}_{-i}$, and $Y$.*

*Proof.* Fix $i$, $\boldsymbol{r}_{-i}$, and $Y$, and write $q_i(r) := q_i((r, \boldsymbol{r}_{-i}), Y)$ and $p_i(r) := p_i((r, \boldsymbol{r}_{-i}), Y)$.

*Necessity.* For $0 \le s < t \le \bar{\theta}$, statewise DSIC gives

$$tu(q_i(t)) - p_i(t) \ge tu(q_i(s)) - p_i(s), \quad su(q_i(s)) - p_i(s) \ge su(q_i(t)) - p_i(t).$$

Rearranging yields

$$(t-s)u(q_i(s)) \le [tu(q_i(t)) - p_i(t)] - [su(q_i(s)) - p_i(s)] \le (t-s)u(q_i(t)).$$

Since $t - s > 0$, these bounds imply that $u(q_i(r))$ is nondecreasing.

For $r > 0$ and a partition $0 = t_0 < \cdots < t_K = r$, summing these bounds over adjacent points gives

$$\sum_{k=1}^{K} u(q_i(t_{k-1}))(t_k - t_{k-1}) \le ru(q_i(r)) - p_i(r) + p_i(0) \le \sum_{k=1}^{K} u(q_i(t_k))(t_k - t_{k-1}).$$

These are lower and upper Riemann sums of a monotone function bounded between zero and $u(\bar{Q})$. Both converge to the same integral as the largest subinterval length tends to zero. Hence

$$ru(q_i(r)) - p_i(r) = -p_i(0) + \int_0^r u(q_i(t))\, dt.$$

The identity also holds at $r = 0$. Rearranging proves the payment formula, and setting $r = \theta_i$ gives truthful utility.

*Sufficiency.* Under the stated monotonicity and payment formula, the gain from truthful reporting rather than reporting $r_i$ is

$$\begin{aligned}
&[\theta_i u(q_i(\theta_i)) - p_i(\theta_i)] - [\theta_i u(q_i(r_i)) - p_i(r_i)] \\
&\qquad = \begin{cases} \displaystyle\int_{r_i}^{\theta_i} [u(q_i(t)) - u(q_i(r_i))]\, dt, & r_i \le \theta_i, \\ \displaystyle\int_{\theta_i}^{r_i} [u(q_i(r_i)) - u(q_i(t))]\, dt, & r_i > \theta_i. \end{cases}
\end{aligned}$$

Both integrands are nonnegative by monotonicity. Since the fixed user, other reports and outside bid were arbitrary, statewise DSIC follows.

*Individual rationality.* Since $u(q_i(t)) \ge 0$, truthful utility is minimised at $\theta_i = 0$, where it equals $-p_i(0)$. Thus EPIR is equivalent to $p_i(0) \le 0$ for every $i, \boldsymbol{r}_{-i}, Y$, as stated. □

## A.3 BIC characterisation

**Proposition A.7.** *Suppose interim expected payments are finite. A pair of bid and allocation rules $(B, \boldsymbol{q})$ is BIC-implementable if and only if, for every user $i$, $\mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((r_i, \boldsymbol{\theta}_{-i}), Y))\right]$ is nondecreasing in $r_i$. Every BIC implementation satisfies*

$$U_i(\theta_i) = U_i(0) + \int_0^{\theta_i} \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))\right]\, dt,$$

*and*

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[p_i((\theta_i, \boldsymbol{\theta}_{-i}), Y)\right] = \theta_i \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((\theta_i, \boldsymbol{\theta}_{-i}), Y))\right] \\ - \int_0^{\theta_i} \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))\right]\, dt - U_i(0).\end{aligned}$$

*Whenever $(B, \boldsymbol{q})$ admits a BIC implementation, define the payment rule by*

$$\begin{aligned}p_i^0((r_i, \boldsymbol{r}_{-i}), Y) = \mathbb{1}\{B((r_i, \boldsymbol{r}_{-i})) \geq Y\} \\ \times \frac{r_i \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((r_i, \boldsymbol{\theta}_{-i}), Y))\right] - \int_0^{r_i} \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))\right]\, dt}{\Pr_{\boldsymbol{\theta}_{-i},Y}\left\{B((r_i, \boldsymbol{\theta}_{-i})) \geq Y\right\}},\end{aligned}$$

*where the ratio is defined to be zero whenever its denominator is zero. This payment rule is bounded, BIC, interim IR, and normalised, and satisfies $p_i^0(\boldsymbol{r}, Y) = 0$ whenever $B(\boldsymbol{r}) < Y$. It yields*

$$U_i^0(\theta_i) = \int_0^{\theta_i} \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}\left[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))\right]\, dt, \qquad U_i^0(0) = 0.$$

*Proof.* Fix a user $i$. By independence, the distribution of $(\boldsymbol{\theta}_{-i}, Y)$ is the same for every type $\theta_i$.

*Necessity.* Suppose the mechanism is BIC. For $0 \leq s < t \leq \bar{\theta}$, rearranging the two BIC constraints gives

$$\begin{aligned}(t-s)\mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[u(q_i((s, \boldsymbol{\theta}_{-i}), Y))] \leq U_i(t) - U_i(s) \\ \leq (t-s)\mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))]\,.\end{aligned}$$

Expected $u(q_i)$ is nondecreasing and lies in $[0, u(\bar{Q})]$. The partition argument in the proof of Proposition A.6 applies to these bounds, giving the stated envelope formula, including at $\theta_i = 0$. The definition of $U_i$ gives the expected-payment formula.

*Sufficiency and the payment.* Conversely, suppose the stated monotonicity holds for every

user and use $p_i^0$ from the proposition. Monotonicity, $0 \le u(q_i) \le u(\bar{Q})$, and zero allocation after a loss give

$$\begin{aligned} 0 &\le r_i \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[u(q_i((r_i,\boldsymbol{\theta}_{-i}),Y))] - \int_0^{r_i} \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[u(q_i((t,\boldsymbol{\theta}_{-i}),Y))]\, dt \\ &\le r_i \mathbb{E}_{\boldsymbol{\theta}_{-i},Y}[u(q_i((r_i,\boldsymbol{\theta}_{-i}),Y))] \\ &\le \bar{\theta} u(\bar{Q}) \Pr_{\boldsymbol{\theta}_{-i},Y}\{B((r_i,\boldsymbol{\theta}_{-i})) \ge Y\}. \end{aligned}$$

If the winning probability is zero, these bounds force the numerator to be zero, and $p_i^0 = 0$ by convention. Otherwise, dividing by the winning probability and multiplying by the winning indicator gives

$$0 \le p_i^0((r_i,\boldsymbol{r}_{-i}),Y) \le \bar{\theta} u(\bar{Q}).$$

Thus payments are bounded and integrable. The winning indicator ensures zero payment after a loss, and the zero numerator at $r_i = 0$ gives $p_i^0((0,\boldsymbol{r}_{-i}),Y) = 0$ in every state, so the rule is normalised.

Taking expectations of the winning indicator cancels any positive denominator. If it is zero, both the numerator and expected payment are zero. Thus the expected-payment formula in the proposition holds for $p_i^0$ with $U_i^0(0) = 0$, for every report $r_i$.

*Incentives and participation.* The integral comparison in the sufficiency proof of Proposition A.6 applies to expected $u(q_i)$, which is nondecreasing in the report. Thus the gain from truthfulness is nonnegative for reports on either side of $\theta_i$. Applying the construction to every user gives a BIC implementation. The expected-payment identity also gives the stated truthful utility, with $U_i^0(\theta_i) \ge 0$ and $U_i^0(0) = 0$, proving interim IR. □

# B Proofs for Section 5

## B.1 Proof of Lemma 5.1

*Proof.* Under truthful reporting, for every state $(\boldsymbol{\theta}, Y)$,

$$\begin{aligned} &\sum_{i=1}^n [\theta_i u(q_i(\boldsymbol{\theta},Y)) - p_i(\boldsymbol{\theta},Y)] + \sum_{i=1}^n p_i(\boldsymbol{\theta},Y) - \mathbb{1}\{B(\boldsymbol{\theta}) \ge Y\}Y \\ &\qquad = \sum_{i=1}^n \theta_i u(q_i(\boldsymbol{\theta},Y)) - \mathbb{1}\{B(\boldsymbol{\theta}) \ge Y\}Y. \end{aligned}$$

Use values are bounded, $\mathbb{E}[Y] < \infty$, and user payments are integrable. Taking expectations and applying the law of iterated expectations to each user's utility therefore gives

$$\Pi(\mathcal{M}) = S(B, \boldsymbol{q}) - \sum_{i=1}^{n} \mathbb{E}_{\theta_i}[U_i(\theta_i)],$$

which proves the lemma. □

## B.2 Proof of Proposition 5.2

**Lemma B.1.** *For any BIC-implementable pair $(B, \boldsymbol{q})$, the upper bound $\bar{\Pi}(B, \boldsymbol{q})$ is attained by a normalised BIC and interim-IR implementation with bounded payments and no transfers after a loss.*

*Proof. Expected information rents.* Fix a BIC and interim-IR mechanism $\mathcal{M} = (B, \boldsymbol{q}, \boldsymbol{p})$ with integrable payments. The envelope integrand in Proposition A.7 is nonnegative and bounded by $u(\bar{Q})$. Averaging over $\theta_i$ and applying Tonelli's theorem gives

$$\begin{aligned}
\mathbb{E}_{\theta_i}[U_i(\theta_i) - U_i(0)] &= \int_0^{\bar{\theta}} \left[ \int_0^{\theta_i} \mathbb{E}_{\boldsymbol{\theta}_{-i}, Y}[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))]\, dt \right] h(\theta_i)\, d\theta_i \\
&= \int_0^{\bar{\theta}} (1 - H(t)) \mathbb{E}_{\boldsymbol{\theta}_{-i}, Y}[u(q_i((t, \boldsymbol{\theta}_{-i}), Y))]\, dt \\
&= \mathbb{E}_{\boldsymbol{\theta}, Y}\left[ \frac{1 - H(\theta_i)}{h(\theta_i)} u(q_i(\boldsymbol{\theta}, Y)) \right].
\end{aligned}$$

The last equality uses $h(t) > 0$ and independence to write the integral as a joint expectation.

*Budget upper bound.* Lemma 5.1 and $\phi(\theta) = \theta - (1 - H(\theta))/h(\theta)$ now give

$$\begin{aligned}
\Pi(\mathcal{M}) &= S(B, \boldsymbol{q}) - \sum_{i=1}^{n} \mathbb{E}_{\theta_i}[U_i(\theta_i) - U_i(0)] - \sum_{i=1}^{n} U_i(0) \\
&= \mathbb{E}\left[ \sum_{i=1}^{n} \phi(\theta_i) u(q_i(\boldsymbol{\theta}, Y)) - \mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\} Y \right] - \sum_{i=1}^{n} U_i(0) \\
&= \bar{\Pi}(B, \boldsymbol{q}) - \sum_{i=1}^{n} U_i(0) \leq \bar{\Pi}(B, \boldsymbol{q}),
\end{aligned}$$

where interim IR gives $U_i(0) \geq 0$. The expected information rents, and hence $\bar{\Pi}(B, \boldsymbol{q})$, depend only on the fixed bid and allocation rules.

*Attainment.* Proposition A.7 supplies bounded, normalised BIC and interim-IR payments $p_i^0$, with no transfers after a loss. Since $U_i^0(0) = 0$, they attain expected budget $\bar{\Pi}(B, \boldsymbol{q})$.

The budget identity gives the same conclusion for every normalised BIC and interim-IR implementation with integrable payments. □

*Proof of Proposition 5.2. General cost-recovery criterion.* For any BIC-implementable pair $(B, \boldsymbol{q})$, Lemma B.1 bounds the expected budget of every BIC and interim-IR implementation with integrable payments by $\bar{\Pi}(B, \boldsymbol{q})$ and provides a normalised implementation attaining it. Hence the pair admits a BIC and interim-IR implementation satisfying expected cost recovery if and only if $\bar{\Pi}(B, \boldsymbol{q}) \geq 0$.

*First-best rules.* Substituting the first-best bid and allocation into $\bar{\Pi}(B, \boldsymbol{q})$, with $u(0) = 0$, gives $\bar{\Pi}^{\circ}$. Normalised VCG implements these rules, is statewise DSIC and EPIR, and has bounded payments. It is therefore BIC and interim IR with integrable payments, and normalisation gives $U_i(0) = 0$. Lemma B.1 yields

$$\mathbb{E}\left[\sum_{i=1}^{n} \widetilde{p}_i(\boldsymbol{\theta}, Y) - \mathbb{1}\{W(\boldsymbol{\theta}) \geq Y\}Y\right] = \bar{\Pi}^{\circ}.$$

Thus normalised VCG attains the largest expected budget for the first-best decisions, and the general criterion gives the stated equivalence. □

## B.3 Proof of Proposition 5.4

*Proof.* Fix $\alpha \in [0, 1]$.

*The weighted-rule bound.* For any BIC-implementable pair $(B, \boldsymbol{q})$, the definitions give

$$L(B, \boldsymbol{q}; \alpha) = \mathbb{E}\left[\sum_{i=1}^{n} \psi(\theta_i; \alpha) u(q_i(\boldsymbol{\theta}, Y)) - \mathbb{1}\{B(\boldsymbol{\theta}) \geq Y\}Y\right]. \tag{21}$$

Fix $\boldsymbol{\theta}$, and abbreviate $\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha)$ as $\widehat{\boldsymbol{\psi}}$. Replacing negative weights by zero and maximising over feasible allocations gives

$$\sum_{i=1}^{n} \psi(\theta_i; \alpha) u(q_i) \leq \sum_{i=1}^{n} \widehat{\psi}(\theta_i; \alpha) u(q_i) \leq W(\widehat{\boldsymbol{\psi}}).$$

The integrand in (21) is therefore at most $W(\widehat{\boldsymbol{\psi}}) - Y$ after a win and zero after a loss. Hence

$$L(B, \boldsymbol{q}; \alpha) \leq \mathbb{E}\left[\max\left\{W\left(\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta}; \alpha)\right) - Y,\, 0\right\}\right]. \tag{22}$$

*Incentives and payments.* Assumption 5.3 makes $\psi(\theta; \alpha) = (1 - \alpha)\theta + \alpha\phi(\theta)$ weakly increasing in $\theta$, and truncation preserves this property.

Fix $i$, $\boldsymbol{r}_{-i}$, and $Y$. For $s < t$, monotonicity of $W$ implies that a win at $s$ remains a win at $t$. Conditional on a win, Lemma A.2 makes user $i$'s allocation nondecreasing in her own adjusted weight. Since her allocation is zero after a loss,

$$\widehat{q}_i((t, \boldsymbol{r}_{-i}), Y; \alpha) \geq \widehat{q}_i((s, \boldsymbol{r}_{-i}), Y; \alpha).$$

Since $u$ is increasing, Proposition A.6 implies that the normalised payment rule in (13) makes the mechanism statewise DSIC and EPIR. Monotonicity and nonnegativity of $u(\widehat{q}_i)$ also give

$$0 \leq \widehat{p}_i(\boldsymbol{r}, Y; \alpha) \leq r_i u(\widehat{q}_i(\boldsymbol{r}, Y; \alpha)) \leq \bar{\theta} u(\bar{Q}).$$

Thus payments are bounded and integrable, and vanish when $r_i = 0$ or $\widehat{q}_i(\boldsymbol{r}, Y; \alpha) = 0$, including after a loss.

*Attainment.* Lemma A.1 and the convention $\boldsymbol{q}^*(\mathbf{0}) = \mathbf{0}$ give $q_i^*(\widehat{\boldsymbol{\psi}}) = 0$ whenever $\widehat{\psi}(\theta_i; \alpha) = 0$. Hence

$$\sum_{i=1}^{n} \psi(\theta_i; \alpha) u\Big(q_i^*(\widehat{\boldsymbol{\psi}})\Big) = \sum_{i=1}^{n} \widehat{\psi}(\theta_i; \alpha) u\Big(q_i^*(\widehat{\boldsymbol{\psi}})\Big) = W(\widehat{\boldsymbol{\psi}}). \tag{23}$$

Since $\widehat{B} = W(\widehat{\boldsymbol{\psi}})$, acquisition occurs exactly when this value is at least $Y$. Substitution into (21) therefore gives equality in (22). The payment construction makes the rules BIC-implementable, so they attain the maximum. Normalisation gives $U_i(0) = 0$, and Lemma B.1 gives expected budget $\bar{\Pi}(\widehat{B}, \widehat{\boldsymbol{q}})$.

*Comparison with arbitrary mechanisms.* For any BIC and interim-IR mechanism $\mathcal{M} = (B, \boldsymbol{q}, \boldsymbol{p})$ with integrable payments, Lemma B.1 and $U_i(0) \geq 0$ give

$$(1-\alpha) S(B, \boldsymbol{q}) + \alpha \Pi(\mathcal{M}) = L(B, \boldsymbol{q}; \alpha) - \alpha \sum_{i=1}^{n} U_i(0) \leq L(B, \boldsymbol{q}; \alpha). \tag{24}$$

Optimality of the constructed rules and attainment of their budget bound therefore imply

$$(1-\alpha) S(\mathcal{M}) + \alpha \Pi(\mathcal{M}) \leq (1-\alpha) S(\alpha) + \alpha \Pi(\alpha). \tag{25}$$

Thus the mechanism also maximises the weighted sum of surplus and actual expected budget over all BIC and interim-IR mechanisms. □

## B.4 Proof of Proposition 5.5

*Proof. Monotonicity.* Fix $0 \leq \alpha_1 < \alpha_2 \leq 1$, and write $S_j = S(\alpha_j)$ and $\Pi_j = \Pi(\alpha_j)$ for $j = 1, 2$. Optimality at each weight and attainment of the budget bounds, established in

Proposition 5.4, give

$$(1-\alpha_1)S_1 + \alpha_1\Pi_1 \geq (1-\alpha_1)S_2 + \alpha_1\Pi_2,$$
$$(1-\alpha_2)S_2 + \alpha_2\Pi_2 \geq (1-\alpha_2)S_1 + \alpha_2\Pi_1.$$

Multiplying these inequalities by $1-\alpha_2$ and $1-\alpha_1$, respectively, and adding cancels the surplus terms. Multiplying by $\alpha_2$ and $\alpha_1$ instead cancels the budget terms. Thus

$$(\alpha_2-\alpha_1)(\Pi_2-\Pi_1) \geq 0, \qquad (\alpha_2-\alpha_1)(S_1-S_2) \geq 0.$$

Since $\alpha_2 > \alpha_1$, $\Pi$ is weakly increasing and $S$ is weakly decreasing on $[0,1]$.

*Continuity of realised allocations.* Let $\alpha_k \to \alpha$ in $[0,1]$. Continuity of adjusted productivity, truncation, and $W$ gives, for every type profile, $\widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha_k) \to \widehat{\boldsymbol{\psi}}(\boldsymbol{\theta};\alpha)$ and $\widehat{B}(\boldsymbol{\theta};\alpha_k) \to \widehat{B}(\boldsymbol{\theta};\alpha)$. Independence and the continuous distribution of $Y$ imply that $Y = \widehat{B}(\boldsymbol{\theta};\alpha)$ has probability zero. Outside this event, the auction outcome is eventually unchanged.

If $Y > \widehat{B}(\boldsymbol{\theta};\alpha)$, the limiting and eventual allocations are zero. If $Y < \widehat{B}(\boldsymbol{\theta};\alpha)$, the group eventually wins. The limiting adjusted profile is then nonzero because $Y \geq 0$, so Lemma A.2 gives continuity of $\boldsymbol{q}^*$ there. A zero limiting adjusted profile has bid zero and lies in the losing case almost surely, because $Y > 0$ almost surely. Therefore

$$\widehat{\boldsymbol{q}}(\boldsymbol{\theta}, Y; \alpha_k) \to \widehat{\boldsymbol{q}}(\boldsymbol{\theta}, Y; \alpha), \quad \mathbb{1}\{\widehat{B}(\boldsymbol{\theta};\alpha_k) \geq Y\}Y \to \mathbb{1}\{\widehat{B}(\boldsymbol{\theta};\alpha) \geq Y\}Y$$

almost surely.

*Continuity of expected surplus and budget.* By the definition of surplus, normalisation, and Lemma B.1, the integrands for $S(\alpha_k)$ and $\Pi(\alpha_k)$ use weights $\theta_i$ and $\phi(\theta_i)$, respectively, and subtract the same acquisition payment. Both converge almost surely by the preceding result and continuity of $u$. Since $h$ is continuous and strictly positive on $[0,\bar{\theta}]$, $\phi$ is bounded. Together with $0 \leq \widehat{q}_i \leq \bar{Q}$, this bounds the absolute values of both integrands, uniformly over all weights, by

$$nu(\bar{Q})\left(\bar{\theta} + \sup_{\theta\in[0,\bar{\theta}]} |\phi(\theta)|\right) + Y.$$

The bound is integrable because $\mathbb{E}[Y] < \infty$. Dominated convergence gives $S(\alpha_k) \to S(\alpha)$ and $\Pi(\alpha_k) \to \Pi(\alpha)$, proving continuity on $[0,1]$. □

## B.5 Proof of Theorem 5.6

*Proof. Existence and feasibility of $\alpha^*$.* By Proposition 5.5, $\Pi$ is continuous. Since $\Pi(1) \geq 0$, the set defining $\alpha^*$ is a nonempty closed subset of $[0,1]$, so its minimum exists and $\Pi(\alpha^*) \geq 0$. Proposition 5.4 gives BIC, interim IR, and integrable payments, so $\widehat{\mathcal{M}}(\alpha^*)$ is feasible.

*Part (i).* If $\bar{\Pi}^\circ = \Pi(0) \geq 0$, then $\alpha^* = 0$. The mechanism $\widehat{\mathcal{M}}(0)$ implements the first-best decisions, which maximise expected surplus even without the budget constraint. It therefore solves the constrained problem.

*Part (ii).* Suppose $\Pi(0) = \bar{\Pi}^\circ < 0$. We first show that $\Pi(1) > 0$. Since there are no transfers after a loss, a negative first-best expected budget requires acquisition with positive probability. The bound $W(\boldsymbol{\theta}) \leq W(\bar{\theta}, \ldots, \bar{\theta})$ and the absence of atoms in $Y$ then give

$$\Pr\{Y < W(\bar{\theta}, \ldots, \bar{\theta})\} > 0.$$

Choose $y < W(\bar{\theta}, \ldots, \bar{\theta})$ with $F_Y(y) > 0$.

Since $H(\bar{\theta}) = 1$, we have $\phi(\bar{\theta}) = \bar{\theta}$. Continuity of $\phi$ and $W$ gives $W(\widehat{\boldsymbol{\phi}}(\boldsymbol{\theta})) > y$ when all types are sufficiently close to $\bar{\theta}$. Independent types with strictly positive density assign positive probability to this set. Independence from $Y$ therefore gives

$$\Pr\Big\{W\Big(\widehat{\boldsymbol{\phi}}(\boldsymbol{\theta})\Big) > y,\ Y \leq y\Big\} = \Pr\Big\{W\Big(\widehat{\boldsymbol{\phi}}(\boldsymbol{\theta})\Big) > y\Big\}\, F_Y(y) > 0.$$

The integrand in (15) is nonnegative everywhere and strictly positive on this event, so $\Pi(1) > 0$.

*IVT.* Since $\Pi(0) < 0 < \Pi(1)$, continuity makes weights sufficiently close to zero infeasible and some weights strictly below one feasible. Thus $0 < \alpha^* < 1$. Moreover, $\Pi(\alpha^*) = 0$, since a positive value would make a slightly smaller weight feasible by continuity, contradicting minimality.

*Optimality.* Let $\mathcal{M}$ be any BIC and interim-IR mechanism with integrable payments and $\Pi(\mathcal{M}) \geq 0$. Equation (25) and $\Pi(\alpha^*) = 0$ give

$$(1-\alpha^*)S(\mathcal{M}) \leq (1-\alpha^*)S(\mathcal{M}) + \alpha^*\Pi(\mathcal{M}) \leq (1-\alpha^*)S(\alpha^*).$$

Since $1 - \alpha^* > 0$, division gives $S(\mathcal{M}) \leq S(\alpha^*)$. Thus $\widehat{\mathcal{M}}(\alpha^*)$ is optimal over all feasible mechanisms. □

# C Proofs for Section 6

This appendix proves the allocation ordering used to compare the first best with optimal cost recovery.

## C.1 Proof of Theorem 6.2

**Lemma C.1.** *Let $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathbb{R}^n_+ \setminus \{\mathbf{0}\}$, with $\theta_1 \geq \cdots \geq \theta_n$. Suppose $\theta'_i = \tau_i \theta_i$, with $\tau_1 \geq \cdots \geq \tau_n \geq 0$. Then*

$$\{i : q^*_i(\boldsymbol{\theta}') > 0\} \subseteq \{i : q^*_i(\boldsymbol{\theta}) > 0\}, \quad \sum_{i=1}^{k} q^*_i(\boldsymbol{\theta}') \geq \sum_{i=1}^{k} q^*_i(\boldsymbol{\theta}), \quad k < n.$$

*Thus $\boldsymbol{q}^*(\boldsymbol{\theta}')/\bar{Q}$ majorizes $\boldsymbol{q}^*(\boldsymbol{\theta})/\bar{Q}$.*

*Proof.* Write $\boldsymbol{q} = \boldsymbol{q}^*(\boldsymbol{\theta})$ and $\boldsymbol{q}' = \boldsymbol{q}^*(\boldsymbol{\theta}')$. Lemma A.1 gives uniqueness, $\sum_i q_i = \sum_i q'_i = \bar{Q}$, and capacity multipliers $\lambda, \lambda' \geq 0$.

*Ordering of allocations.* If $i < j$ but $q_i < q_j$, swapping the quantities changes the objective by

$$(\theta_i - \theta_j)\big[u(q_j) - u(q_i)\big] \geq 0.$$

A strict gain contradicts optimality, while equality gives a distinct optimiser and contradicts uniqueness. Thus $q_1 \geq \cdots \geq q_n$. The same argument gives $q'_1 \geq \cdots \geq q'_n$, since $\theta'_i = \tau_i \theta_i$ is also nonnegative and weakly decreasing.

*Single crossing.* Put $\delta_i = q'_i - q_i$. Suppose $\delta_i < 0 < \delta_j$ for some $i < j$. The positive allocations $q_i$ and $q'_j$ imply $\theta_i > 0$ and $\theta'_j > 0$. Hence $\theta_j > 0$, $\tau_i \geq \tau_j > 0$, and $\theta'_i > 0$.

At zero, interpret the derivative as $u'(0+)$. If this is infinite, the positive weights $\theta'_i$ and $\theta_j$ imply $q'_i > 0$ and $q_j > 0$, by Lemma A.1. Otherwise the KKT boundary inequalities apply. Since $q'_i < q_i$ and $q_j < q'_j$, the KKT conditions and strict decrease of $u'$ give

$$\lambda' \geq \theta'_i u'(q'_i) > \theta'_i u'(q_i) = \tau_i \lambda, \quad \tau_j \lambda \geq \theta'_j u'(q_j) > \theta'_j u'(q'_j) = \lambda'.$$

Since $\tau_i \geq \tau_j$ and $\lambda \geq 0$, including $\lambda = 0$, these imply the contradiction $\lambda' > \tau_i \lambda \geq \tau_j \lambda > \lambda'$. Thus a negative $\delta_i$ cannot precede a positive one.

*Majorization and inclusion.* Since $\sum_i \delta_i = 0$, a negative partial sum at $k < n$ would require $\delta_i < 0 < \delta_j$ for some $i \leq k < j$. Hence

$$\sum_{i=1}^{k} q'_i \geq \sum_{i=1}^{k} q_i, \qquad k = 1, \ldots, n-1.$$

Together with the common ordering and equal totals, this proves majorization.

Let $\ell$ be the number of positive components of $\boldsymbol{q}$. If $\ell = n$, inclusion is immediate. Otherwise, ordering gives $q_i > 0$ exactly for $i \le \ell$, and the inequality at $k = \ell$ gives

$$\bar{Q} = \sum_{i=1}^{\ell} q_i \le \sum_{i=1}^{\ell} q'_i \le \bar{Q}.$$

Nonnegativity and $\sum_i q'_i = \bar{Q}$ force $q'_i = 0$ for $i > \ell$, proving the active-set inclusion. □

*Proof of Theorem 6.2. Ordering of adjusted weights.* Since $\alpha_2 \ge \alpha_1$, adjusted productivity satisfies $0 \le \widehat{\psi}(\theta_i; \alpha_2) \le \widehat{\psi}(\theta_i; \alpha_1)$ for every $i$. If the adjusted profile at $\alpha_2$ is zero, the convention $\boldsymbol{q}^*(\mathbf{0}) = \mathbf{0}$ gives part (i), while part (ii) does not apply. Otherwise, both profiles are nonzero. Regularity and $\theta_1 \ge \cdots \ge \theta_n$ imply that the adjusted weights at $\alpha_1$ are weakly decreasing in $i$, with positive weights preceding any zeros.

*Scaling factors and their monotonicity.* If $\widehat{\psi}(\theta; \alpha_1) > 0$, then $\theta > 0$ and $1 - \alpha_1 \chi(\theta) > 0$. Cancelling the common factor $\theta$ gives

$$\frac{\widehat{\psi}(\theta; \alpha_2)}{\widehat{\psi}(\theta; \alpha_1)} = \max\left\{ \frac{1 - \alpha_2 \chi(\theta)}{1 - \alpha_1 \chi(\theta)}, 0 \right\}. \tag{26}$$

Where the denominator is positive, the derivative of the untruncated fraction is

$$\frac{d}{d\chi}\left[\frac{1 - \alpha_2 \chi}{1 - \alpha_1 \chi}\right] = \frac{\alpha_1 - \alpha_2}{[1 - \alpha_1 \chi]^2} \le 0.$$

Taking the maximum with zero preserves this weakly decreasing dependence on $\chi$. For $i < j$ with $\widehat{\psi}(\theta_j; \alpha_1) > 0$, both denominators are positive and Assumption 6.1 gives $\chi(\theta_i) \le \chi(\theta_j)$. Thus the scaling factors in (26) are nonnegative and weakly decreasing in the user index.

*Majorization and inclusion.* A zero adjusted weight at $\alpha_1$ remains zero at $\alpha_2$. Setting its scaling factor to zero extends the preceding factors without changing their weakly decreasing order. Lemma C.1 therefore applies to the two nonzero adjusted profiles and gives the active-set inclusion and all the stated top-$k$ inequalities. □

# D Proofs for Section 7

Couple all group sizes by an infinite sequence of independent types with distribution $H$, independent of the same outside bid $Y$. The $m$-user economy contains the first $m$ draws, and $W_0 = S_0^\circ = 0$.

## D.1 Proof of Proposition 7.1

*Proof.* Fix $m \geq 1$. By (5), normalised VCG leaves user $i$ with her marginal contribution to first-best surplus:

$$\max\{W_m(\boldsymbol{\theta}) - Y, 0\} - \max\{W_{m,-i}(\boldsymbol{\theta}_{-i}) - Y, 0\}.$$

The remaining users can reallocate the full capacity $\bar{Q}$. Removing a fixed user leaves $m-1$ independent draws from $H$, independent of the same outside bid $Y$. Taking expectations therefore gives

$$\mathbb{E}_{\theta_i}[U_i(\theta_i)] = S^\circ_m - S^\circ_{m-1}.$$

Normalisation gives $U_i(0) = 0$, so these expected utilities are also information rents, and Lemma B.1 implies that VCG attains $\bar{\Pi}^\circ_m$. Summing users' rents in the surplus–budget identity yields

$$\bar{\Pi}^\circ_m = S^\circ_m - \sum_{i=1}^{m} \mathbb{E}_{\theta_i}[U_i(\theta_i)] = S^\circ_m - m\Big(S^\circ_m - S^\circ_{m-1}\Big),$$

proving (19). For $m = 1$, $S^\circ_0 = 0$ gives $\bar{\Pi}^\circ_1 = 0$. The cost-recovery equivalence in (20) follows by rearrangement. For $m \geq 2$, the identity also gives

$$\bar{\Pi}^\circ_m = m(m-1)\left[\frac{S^\circ_{m-1}}{m-1} - \frac{S^\circ_m}{m}\right],$$

establishing the equivalent comparison of average surplus stated in the text. □

## D.2 Proof of Theorem 7.2

**Lemma D.1.** *The sequence $(\mathbb{E}[W_m])_{m\geq 0}$ is nondecreasing and has nonincreasing first differences.*

*Proof. Group value representation.* For a finite set $T$ of users with fixed types, let $W(T)$ be its optimised use value, with $W(\varnothing) = 0$, and define

$$d_i(t) = \int_0^{\bar{Q}} \mathbb{1}\{\theta_i u'(q) > t\}\, dq, \qquad t \geq 0.$$

The identity $u(q) = \int_0^q u'(s)\, ds$ holds even when the integral is improper at zero. Values at $q = 0$ do not affect these integrals, and $d_i(t) = 0$ for zero types. Since marginal benefits

decrease with capacity, Tonelli's theorem gives, for every feasible allocation,

$$\sum_{i\in T}\theta_i u(q_i)=\int_0^\infty \sum_{i\in T}\min\{q_i,d_i(t)\}\,dt\le\int_0^\infty \min\left\{\bar{Q},\sum_{i\in T}d_i(t)\right\}\,dt.$$

*Attainment.* If $T$ is empty or all types are zero, both sides are zero. Otherwise, let $q_i^*$ and $\lambda\ge 0$ be the optimal allocation and capacity multiplier from Lemma A.1. The KKT conditions, including the boundary inequalities, give

$$d_i(t)\ge q_i^* \quad (0\le t<\lambda),\qquad d_i(t)\le q_i^* \quad (t>\lambda).$$

Together with $\sum_{i\in T}q_i^*=\bar{Q}$, this makes the integrands in the preceding bound equal for every $t\neq\lambda$. Hence

$$W(T)=\int_0^\infty \min\left\{\bar{Q},\sum_{i\in T}d_i(t)\right\}\,dt.$$

*Marginal gains.* For $e\notin T$, the representation gives

$$W(T\cup\{e\})-W(T)=\int_0^\infty \min\left\{d_e(t),\max\left\{\bar{Q}-\sum_{i\in T}d_i(t),0\right\}\right\}\,dt.$$

The integrand is nonnegative and weakly decreases as $T$ expands. Thus, for $T\subseteq T'$ and $e\notin T'$,

$$W(T\cup\{e\})-W(T)\ge W(T'\cup\{e\})-W(T')\ge 0.$$

*Expectations.* For $m\ge 1$, take $T=\{1,\dots,m-1\}$, empty when $m=1$, and compare adding user $m+1$ before and after user $m$. The two sets of $m$ users have the same distribution. Taking expectations gives

$$\mathbb{E}[W_m]-\mathbb{E}[W_{m-1}]\ge\mathbb{E}[W_{m+1}]-\mathbb{E}[W_m]\ge 0,$$

proving both claims. □

**Lemma D.2.** *Let $v_0=0$, and suppose $(v_m)$ is nondecreasing with nonincreasing first differences.*

*(i) If $(v_m)$ is bounded, then $m(v_m-v_{m-1})\to 0$.*

*(ii) If $v_m\to\infty$ and $v_m/m\to 0$, then $v_m-m(v_m-v_{m-1})\to\infty$.*

*Proof.* Write $d_m=v_m-v_{m-1}$, so $d_m\ge 0$ is nonincreasing and $v_m=\sum_{j=1}^m d_j$.

*Part (i).* There are at least $m/2$ increments after index $\lfloor m/2 \rfloor$, each at least $d_m$. Thus

$$0 \le \frac{m}{2} d_m \le v_m - v_{\lfloor m/2 \rfloor} \to 0,$$

because both terms on the right converge to the same finite limit. Hence $m d_m \to 0$.

*Part (ii).* Nonincreasing increments give $0 \le d_m \le v_m/m \to 0$. For every fixed $k \ge 1$ and $m \ge k$,

$$v_m - m d_m = \sum_{j=1}^{m} (d_j - d_m) \ge v_k - k d_m.$$

Hence $\liminf_{m\to\infty}(v_m - m d_m) \ge v_k$. Letting $k \to \infty$ proves the claim. □

*Proof of Theorem 7.2.* Proposition 7.1 gives

$$\bar{\Pi}^\circ_m = S^\circ_m - m\Big(S^\circ_m - S^\circ_{m-1}\Big).$$

Since the new user can receive zero capacity, $W_m \ge W_{m-1}$. For the same outside bid $Y$,

$$0 \le \max\{W_m - Y, 0\} - \max\{W_{m-1} - Y, 0\} \le W_m - W_{m-1}.$$

Taking expectations gives

$$0 \le S^\circ_m - S^\circ_{m-1} \le \mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]. \tag{27}$$

By Lemma D.1, $(\mathbb{E}[W_m])_{m\ge 0}$ is nondecreasing with nonincreasing first differences. We apply the two parts of Lemma D.2 to this sequence.

*Finite marginal benefit at zero.* Suppose $u'(0+) < \infty$. Concavity and $u(0) = 0$ give $u(q) \le u'(0+)q$, so bounded types and the capacity constraint imply $W_m \le \bar{\theta} u'(0+)\bar{Q}$. Thus $\mathbb{E}[W_m]$ is bounded. Lemma D.2(i) and (27) give

$$0 \le m\Big(S^\circ_m - S^\circ_{m-1}\Big) \le m\Big(\mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]\Big) \to 0.$$

Hence aggregate expected information rents vanish.

If $S^\circ_{m_0} > 0$ for some $m_0$, monotonicity gives $S^\circ_m \ge S^\circ_{m_0}$ for $m \ge m_0$. For all sufficiently large $m$,

$$\bar{\Pi}^\circ_m \ge S^\circ_{m_0} - m\Big(S^\circ_m - S^\circ_{m-1}\Big) > 0.$$

If no group size has positive surplus, $S^\circ_m = 0$ for every $m$, so the budget identity gives $\bar{\Pi}^\circ_m = 0$ for every $m$.

*Infinite marginal benefit at zero.* Suppose $u'(0+) = \infty$. Equal division gives the lower bound below. Bounded types, Jensen's inequality, and the capacity constraint give the upper bound:

$$u(\bar{Q}/m)\sum_{i=1}^{m}\theta_i \le W_m \le \bar{\theta}mu(\bar{Q}/m).$$

Taking expectations yields

$$\mathbb{E}[\theta]mu(\bar{Q}/m) \le \mathbb{E}[W_m] \le \bar{\theta}mu(\bar{Q}/m), \tag{28}$$

where $\mathbb{E}[\theta] > 0$ because strictly positive types have positive probability.

Concavity and $u(0) = 0$ imply $u(q)/q \ge u'(q) \to \infty$ as $q \to 0$. Hence

$$mu(\bar{Q}/m) = \bar{Q}\frac{u(\bar{Q}/m)}{\bar{Q}/m} \to \infty.$$

The lower bound in (28) gives $\mathbb{E}[W_m] \to \infty$. The upper bound and continuity of $u$ at zero give

$$0 \le \frac{\mathbb{E}[W_m]}{m} \le \bar{\theta}u(\bar{Q}/m) \to 0.$$

Lemma D.2(ii) therefore yields

$$\mathbb{E}[W_m] - m\Big(\mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]\Big) \to \infty.$$

Since $\max\{W_m - Y, 0\} \ge W_m - Y$, we have $S_m^\circ \ge \mathbb{E}[W_m] - \mathbb{E}[Y]$. Combining this with (27) gives

$$\bar{\Pi}_m^\circ = S_m^\circ - m\Big(S_m^\circ - S_{m-1}^\circ\Big) \ge \mathbb{E}[W_m] - m\Big(\mathbb{E}[W_m] - \mathbb{E}[W_{m-1}]\Big) - \mathbb{E}[Y] \to \infty,$$

because $\mathbb{E}[Y] < \infty$ is fixed.

*Feasibility and optimality.* Normalised VCG is statewise DSIC and EPIR, and the preceding arguments give a nonnegative expected budget for all sufficiently large $m$. It is therefore feasible and attains first-best surplus, solving the design problem at those group sizes. □

## D.3 Proof of Theorem 7.3

Throughout this subsection, $q_i^*$ denotes user $i$'s first-best internal allocation in the $m$-user economy. Dependence on the realised types is suppressed.

**Lemma D.3.** *Suppose $u'(0+) = \infty$. Under the common coupling introduced at the start of this appendix:*

*(i) The largest individual capacity share vanishes:*

$$\max_{i \le m} \frac{q_i^*}{\bar{Q}} \to 0 \qquad \textit{almost surely.} \tag{29}$$

*(ii) Group value diverges even after any one user is removed:*

$$W_m \to \infty, \qquad \min_{i \le m} W_{m,-i} \to \infty \qquad \textit{almost surely.}$$

*Proof.* Since $H$ has no atom at zero, all types in the countable sequence are positive on an event of probability one. Work on this event and the probability-one events for the strong-law limits below.

*Part (i).* Let $\lambda_m$ be the capacity multiplier in the $m$-user first-best allocation. Lemma A.1 gives

$$q_i^* > 0, \qquad \theta_i u'(q_i^*) = \lambda_m, \qquad \sum_{i=1}^{m} q_i^* = \bar{Q}.$$

For $N_m = \sum_{i=1}^{m} \mathbb{1}\{\theta_i \ge \bar{\theta}/2\}$, the strong law gives

$$\frac{N_m}{m} \to 1 - H(\bar{\theta}/2) > 0.$$

Thus $N_m \to \infty$. Eventually, at least one of these users receives at most $\bar{Q}/N_m$, because their total allocation is at most $\bar{Q}$. For such a user, decreasing marginal benefits give

$$\lambda_m = \theta_i u'(q_i^*) \ge \frac{\bar{\theta}}{2} u'(\bar{Q}/N_m) \to \infty.$$

For any fixed $\varepsilon \in (0, \bar{Q})$, a user with $q_i^* \ge \varepsilon$ would instead satisfy

$$\lambda_m = \theta_i u'(q_i^*) \le \bar{\theta} u'(\varepsilon) < \infty.$$

This bound is uniform in $i$ and $m$, so no such user exists for sufficiently large $m$. Since $\varepsilon > 0$ is arbitrary, $\max_{i \le m} q_i^*/\bar{Q} \to 0$.

*Part (ii).* Equal division gives

$$W_m \geq u(\bar{Q}/m) \sum_{i=1}^{m} \theta_i = \bar{Q} \frac{u(\bar{Q}/m)}{\bar{Q}/m} \left( \frac{1}{m} \sum_{i=1}^{m} \theta_i \right) \to \infty.$$

Here $u(q)/q \geq u'(q) \to \infty$ by concavity and $u(0) = 0$, while the sample mean tends to $\mathbb{E}[\theta] > 0$ almost surely. After removing $i$, keeping the remaining users' original allocations gives $W_{m,-i} \geq W_m - \theta_i u(q_i^*)$. Since $\theta_i u(q_i^*) \leq \bar{\theta} u(\bar{Q})$,

$$\min_{i \leq m} W_{m,-i} \geq W_m - \bar{\theta} u(\bar{Q}) \to \infty.$$

□

**Lemma D.4.** *Suppose $u'(0+) = \infty$ and $qu'(q)/u(q) \to \rho \in [0,1]$ as $q \to 0$. For each group size $m \geq 1$, let*

$$C_m = \sum_{i=1}^{m} \left[ W_{m,-i} - \left( W_m - \theta_i u(q_i^*) \right) \right]$$

*be the aggregate within-group externality at the first-best internal allocation. Under the common coupling,*

$$0 \leq C_m \leq W_m \quad \text{for every } m,$$

*and*

$$\frac{C_m}{W_m} \to \rho \qquad \text{almost surely.} \tag{30}$$

*Proof.* For $m = 1$, $W_{1,-1} = 0$ and $W_1 = \theta_1 u(\bar{Q})$, so $C_1 = 0$. We consider $m \geq 2$ below.

*Allocations.* Work on the probability-one event in Lemma D.3 where all types are positive and $\max_{i \leq m} q_i^*/\bar{Q} \to 0$. Lemma A.1 gives

$$0 < q_i^* < \bar{Q}, \qquad \theta_i u'(q_i^*) = \lambda_m, \qquad \sum_{i=1}^{m} q_i^* = \bar{Q},$$

where $\lambda_m$ is the capacity multiplier. These internal allocations are defined whether or not the group acquires.

*Upper bound.* Keeping the remaining users' original allocations is feasible after removing $i$, and any allocation without $i$ is feasible for the full group by giving her zero. Hence

$$W_m - \theta_i u(q_i^*) \leq W_{m,-i} \leq W_m,$$

and subtracting $W_m - \theta_i u(q_i^*)$ and summing gives

$$0 \le C_m \le \sum_{i=1}^{m} \theta_i u(q_i^*) = W_m.$$

For any feasible allocation $(z_j)_{j\neq i}$ after removal, concavity and the KKT conditions give

$$\sum_{j\neq i} \theta_j [u(z_j) - u(q_j^*)] \le \sum_{j\neq i} \theta_j u'(q_j^*)(z_j - q_j^*) = \lambda_m \left[\sum_{j\neq i} z_j - (\bar{Q} - q_i^*)\right] \le \lambda_m q_i^*.$$

The last inequality uses $\sum_{j\neq i} z_j \le \bar{Q}$ and $\lambda_m \ge 0$. Maximising the left-hand side gives user $i$'s within-group externality. Summing over $i$ therefore yields

$$C_m \le \lambda_m \bar{Q} = \sum_{i=1}^{m} \theta_i q_i^* u'(q_i^*) = \sum_{i=1}^{m} \theta_i u(q_i^*) \frac{q_i^* u'(q_i^*)}{u(q_i^*)}.$$

The denominators are positive because $q_i^* > 0$ and $u$ is strictly increasing from zero.

Fix $\varepsilon > 0$. By the elasticity limit and (29), all the displayed ratios are at most $\rho + \varepsilon$ for sufficiently large $m$. Since use values sum to $W_m > 0$,

$$C_m \le (\rho + \varepsilon) W_m, \qquad \frac{C_m}{W_m} \le \rho + \varepsilon. \tag{31}$$

When $\rho = 0$, this and $C_m \ge 0$ already give $C_m / W_m \to 0$.

*Lower bound.* Suppose $0 < \rho \le 1$, and fix $0 < \varepsilon < \rho$. After removing user $i$, consider the proportional redistribution

$$z_j = \frac{q_j^*}{1 - q_i^*/\bar{Q}}, \qquad j \neq i.$$

The denominator is positive, and feasibility follows from

$$\sum_{j\neq i} z_j = \frac{\bar{Q} - q_i^*}{1 - q_i^*/\bar{Q}} = \bar{Q}.$$

Moreover, uniformly over every $i \neq j$,

$$0 < q_j^* \le z_j \le \frac{\max_{k\le m} q_k^*}{1 - \max_{k\le m} q_k^*/\bar{Q}} \to 0.$$

Thus, for sufficiently large $m$, the entire interval $[q_j^*, z_j]$ lies where $su'(s)/u(s) \ge \rho - \varepsilon$,

simultaneously for every removed and remaining user. Integrating gives

$$\log \frac{u(z_j)}{u(q_j^*)} = \int_{q_j^*}^{z_j} \frac{u'(s)}{u(s)}\, ds \geq (\rho - \varepsilon) \int_{q_j^*}^{z_j} \frac{ds}{s} = -(\rho - \varepsilon) \log(1 - q_i^*/\bar{Q}).$$

Exponentiating, multiplying by $\theta_j$, and summing over $j \neq i$ yields

$$W_{m,-i} \geq \sum_{j \neq i} \theta_j u(z_j) \geq (1 - q_i^*/\bar{Q})^{-(\rho-\varepsilon)} \Big( W_m - \theta_i u(q_i^*) \Big).$$

For $0 \leq x < 1$, $(1-x)^{-(\rho-\varepsilon)} - 1 \geq (\rho - \varepsilon)x$. Subtracting $W_m - \theta_i u(q_i^*)$, applying this inequality with $x = q_i^*/\bar{Q}$, and summing over $i$ gives

$$\begin{aligned} C_m &\geq (\rho - \varepsilon) \sum_{i=1}^{m} \frac{q_i^*}{\bar{Q}} \Big( W_m - \theta_i u(q_i^*) \Big) \\ &= (\rho - \varepsilon) \left[ W_m - \sum_{i=1}^{m} \frac{q_i^*}{\bar{Q}} \theta_i u(q_i^*) \right] \\ &\geq (\rho - \varepsilon) \left( 1 - \frac{\max_{k \leq m} q_k^*}{\bar{Q}} \right) W_m. \end{aligned}$$

The last two steps use $\sum_i q_i^*/\bar{Q} = 1$ and $q_i^* \leq \max_{k \leq m} q_k^*$. Together with (31), this gives, for sufficiently large $m$,

$$(\rho - \varepsilon) \left( 1 - \frac{\max_{k \leq m} q_k^*}{\bar{Q}} \right) \leq \frac{C_m}{W_m} \leq \rho + \varepsilon, \qquad 0 < \varepsilon < \rho \leq 1. \tag{32}$$

Since the largest allocation share vanishes,

$$\rho - \varepsilon \leq \liminf_{m \to \infty} \frac{C_m}{W_m} \leq \limsup_{m \to \infty} \frac{C_m}{W_m} \leq \rho + \varepsilon.$$

Letting $\varepsilon \to 0$ proves the result for $0 < \rho \leq 1$, completing the proof. □

**Lemma D.5.** *Let $(Z_m)_{m \geq 1}$ be random variables on the common probability space. If*

$$|Z_m| \leq W_m \quad \text{almost surely for every } m, \qquad \frac{Z_m}{W_m} \to \rho \in [0, 1] \quad \text{almost surely,}$$

*then*

$$\frac{\mathbb{E}[Z_m]}{\mathbb{E}[W_m]} \to \rho.$$

*Proof. Bounds.* The equal-division and Jensen bounds used in (28) do not require $u'(0+) = \infty$.

Since types are positive almost surely and $u$ is strictly increasing from zero, they imply

$$0 < \frac{W_m}{mu(\bar{Q}/m)} \leq \bar{\theta}, \qquad \frac{\mathbb{E}[W_m]}{mu(\bar{Q}/m)} \geq \mathbb{E}[\theta] > 0.$$

Both $W_m$ and $Z_m$ are integrable for each $m$, since $|Z_m| \leq W_m$.

*Convergence.* The assumed limit and the upper bound give

$$\frac{|Z_m - \rho W_m|}{mu(\bar{Q}/m)} = \left|\frac{Z_m}{W_m} - \rho\right| \frac{W_m}{mu(\bar{Q}/m)} \to 0 \quad \text{almost surely.}$$

This quantity is bounded by $2\bar{\theta}$, because $|Z_m| \leq W_m$ and $0 \leq \rho \leq 1$. Dominated convergence therefore makes its expectation tend to zero. The lower bound on $\mathbb{E}[W_m]$ now gives

$$\left|\frac{\mathbb{E}[Z_m]}{\mathbb{E}[W_m]} - \rho\right| \leq \frac{1}{\mathbb{E}[\theta]} \mathbb{E}\left[\frac{|Z_m - \rho W_m|}{mu(\bar{Q}/m)}\right] \to 0.$$

□

*Proof of Theorem 7.3. Finite marginal benefit at zero.* Suppose $u'(0+) < \infty$. $u'(0+)$ is strictly positive, since $u$ is strictly increasing and concave. Both $u'(q)$ and $u(q)/q = (u(q) - u(0))/(q - 0)$ converge to $u'(0+)$ as $q \to 0$, so

$$\frac{qu'(q)}{u(q)} = \frac{u'(q)}{u(q)/q} \to \frac{u'(0+)}{u'(0+)} = 1.$$

Thus $\rho = 1$. The proof of Theorem 7.2 gives $m(S^\circ_m - S^\circ_{m-1}) \to 0$. By assumption, choose $m_0$ with $S^\circ_{m_0} > 0$. Monotonicity of surplus gives $S^\circ_m \geq S^\circ_{m_0}$ for $m \geq m_0$, and hence

$$0 \leq \frac{m(S^\circ_m - S^\circ_{m-1})}{S^\circ_m} \leq \frac{m(S^\circ_m - S^\circ_{m-1})}{S^\circ_{m_0}} \to 0 = 1 - \rho.$$

The budget identity $\bar{\Pi}^\circ_m = S^\circ_m - m(S^\circ_m - S^\circ_{m-1})$ then gives $\bar{\Pi}^\circ_m / S^\circ_m \to 1$.

*Infinite marginal benefit at zero.* For the remainder of the proof, suppose $u'(0+) = \infty$. Let $C_m$ denote the aggregate within-group externality defined in Lemma D.4. That lemma gives

$$\frac{C_m}{W_m} \to \rho \qquad \text{almost surely.}$$

Let

$$\pi_m = \sum_{i=1}^{m} \widetilde{p}_i(\boldsymbol{\theta}, Y) - \mathbb{1}\{W_m \geq Y\} Y$$

be the mediator's realised normalised-VCG budget. Its expectation is $\bar{\Pi}^{\circ}_m$, because normalised VCG attains the largest expected budget compatible with the first-best rules.

The bounds $W_m - \theta_i u(q_i^*) \le W_{m,-i} \le W_m$ established in the proof of Lemma D.4, together with $Y \le W_m$ and (4), give, on a win,

$$0 \le \widetilde{p}_i(\boldsymbol{\theta}, Y) \le \theta_i u(q_i^*).$$

Since use values sum to $W_m$ and $0 \le Y \le W_m$, this implies $|\pi_m| \le W_m$. After a loss, $\pi_m = 0$, so this bound holds in every state.

Lemma D.3(ii) gives $\min_{i \le m} W_{m,-i} \to \infty$ almost surely. Under the common coupling, $Y$ is the same finite random variable at every group size. Therefore, along almost every coupled sample path, there is a size beyond which $W_{m,-i} > Y$ for every user.

At these group sizes the group wins. By (4) and the definition of $C_m$,

$$\pi_m = C_m - Y.$$

Since $C_m/W_m \to \rho$, $W_m \to \infty$, and $Y$ is finite on the sample path,

$$\frac{\pi_m}{W_m} = \frac{C_m}{W_m} - \frac{Y}{W_m} \to \rho \qquad \text{almost surely.}$$

We have established $|\pi_m| \le W_m$ and $\pi_m/W_m \to \rho$ almost surely. Applying Lemma D.5 with $Z_m = \pi_m$, and using $\mathbb{E}[\pi_m] = \bar{\Pi}^{\circ}_m$, gives

$$\frac{\bar{\Pi}^{\circ}_m}{\mathbb{E}[W_m]} \to \rho.$$

It remains to replace expected group value by first-best surplus. Taking expectations of $\max\{W_m - Y, 0\} = W_m - \min\{W_m, Y\}$ gives

$$S^{\circ}_m = \mathbb{E}[W_m] - \mathbb{E}[\min\{W_m, Y\}]. \tag{33}$$

The amount subtracted is uniformly bounded:

$$0 \le \mathbb{E}[\min\{W_m, Y\}] \le \mathbb{E}[Y] < \infty.$$

The proof of Theorem 7.2 gives $\mathbb{E}[W_m] \to \infty$ in this infinite-slope case. Hence

$$\frac{S^{\circ}_m}{\mathbb{E}[W_m]} = 1 - \frac{\mathbb{E}[\min\{W_m, Y\}]}{\mathbb{E}[W_m]} \to 1.$$

In particular, $S_m^\circ > 0$ for all sufficiently large $m$. Combining this with $\bar{\Pi}_m^\circ / \mathbb{E}[W_m] \to \rho$ gives

$$\frac{\bar{\Pi}_m^\circ}{S_m^\circ} = \frac{\bar{\Pi}_m^\circ}{\mathbb{E}[W_m]} \frac{\mathbb{E}[W_m]}{S_m^\circ} \to \rho.$$

Finally, Proposition 7.1 implies

$$\frac{m(S_m^\circ - S_{m-1}^\circ)}{S_m^\circ} = 1 - \frac{\bar{\Pi}_m^\circ}{S_m^\circ} \to 1 - \rho.$$

□

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